

**DESIGN AND IMPLEMENTATION OF A PREDICTIVE MODEL FOR
ANALYZING AND FORECASTING CART ABANDONMENT IN E-COMMERCE
USING THE RANDOM FOREST ALGORITHM**

BY

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CERTIFICATION

This is to certify that this project, DESIGN AND IMPLEMENTATION OF A PREDICTIVE MODEL FOR ANALYZING AND FORECASTING CART ABANDONMENT IN E-COMMERCE USING THE RANDOM FOREST ALGORITHM, was carried out by ONWUKA MELISSA EBERECHI (Matriculation Number: BU22CSC2012) of Computer Science programme under my supervision

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Date

DEDICATION

This study project is dedicated to the Almighty God, who has guided and preserved me throughout my life. I also wish to dedicate it to my mother and my sister, whose love and support have been invaluable, especially in sponsoring this project.

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CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The revolution of the digital world has transformed business in a significant manner and e-commerce has become an important component of the world retail industry. Such a transition has put within easy reach of customers a huge marketplace where they can shop wherever and whenever possible in all kinds of goods and services. The ease, choice and the commonly reduced cost of online shops have contributed to the blistering development of e-commerce making it a crucial mode of communication to companies that wish to gain more customers and serve them as their needs evolve.

Nevertheless, online shopping has its unique problems due to the high speed of online shopping. Cart abandonment is one of the critical issues and happens when a customer puts the product in the shopping cart but fails to make a purchase. The issue, which causes sales and revenues of online businesses of any size to be negatively affected, is that there is a gap between the desire to purchase something and the ability to complete a transaction.

Its impact is high based on the high abandonment of the cart. According to reports of some organizations such as the Baymard Institute (2023), world average cart abandonment rates are dangerously approaching 70 percent. It follows that of ten possible customers who go into an online shop and pick items, approximately 7 will fail to make purchase. Loss of this great deal of leadoffs results in loss of revenues, acquisition of poor marketing results, and may retard the entire growth and rising profits of the business.

Customers abandon their carts due to a variety of reasons, and they include the technical, logistical, economic, and the psychological. Error delays in the checkout process or other technical issues such as web bugs may be frustrating to a user hence making them abandon. The problem with logistics concerning shipping charges, delivery rates and the policies on the returns of the products also de-motivate the buyers. Having to pay a concealed fee or lack of payment choices are also important economic reasons. On top of that all the psychological factors such as worrying about security or general lack of trust to the reliability of the web site also fuel this complicated behaviour.

To be able to overcome the issue of cart abandonment, one should perceive the causes and the ways to forecast customer behaviour or our way of encouraging them to make the purchase. In this respect, online retailers get a powerful tool that is predictive analytics. Predictive analytics involves the past information together with statistical analysis and machine learning algorithms to discover trends and predict the upcoming events. In the case of e-commerce business, including determining who among their customers may abandon their carts. This can be done by examining the different customer interactions on their language, which includes the browsing behaviour, product view, demographics, purchase history and cart behaviour.

Machine learning algorithms, such as the Linear Regression or Random Forest, are especially helpful; in contrast to less sophisticated algorithms, these powerful tools can perform analysis of complicated data sets and reveal our previously unknown associations. You can teach these models historical transaction data, that is all the transactions the business has done, all the sales and the abandoned carts, then, the business can learn the likelihood of an individual shopper abandoning their cart before going on consummating the purchase.

E-commerce business uses machine learning technologies to make pro-active steps to increase sales. Given that such businesses can know customers who are about to walk away without making a purchase in a real-time manner, they could use specialized strategies to win them back to make the possible final purchases. Examples of such strategies can be sending specific personified emails or SMS messages, offering a specific discount or free shipping, simplification of the checkout procedure to a specific user, improving customer support or a fix of specific causes, which may have prompted the customer to abandon their shopping carts. With the adoption of predictive analytics and machine learning instruments, the e-commerce companies can take the challenges proactively to tackling challenges, as opposed to reacting to them. Such a strategy serves to protect the possible income and meet the needs of the customers, though it promotes stable development in the competitive online world. This paper would examine the effectiveness of such predictive approaches in minimizing the occurrence of cart abandonment in e-commerce as well as in insights meant to assist in the formulation of more effective plans.

1.2 STATEMENT OF THE PROBLEM

Over three quarters of online purchasers abandon their shopping cart without paying and this translates to losses in billions of dollars annually. The existing prediction mechanisms of this problem lack precision and are not easily comprehensible and hence it is difficult to act upon them by businesses. Companies require robust predictive models which can identify in real time, the customers who may be at the risk of goods being left in cart and provide helpful insight in this regard so that this situation could be avoided. This research will design a simple and presentable machine learning tool capable of projecting cart abandonment and give businesses an opportunity to respond timely. (Saleh,2022)

1.3 AIM OF THE STUDY

The study shall utilize a predictive analysis model to analyze and determine the reasons that make customers who had initially been planning to make a purchase, abandon their shopping carts whilst performing online check-out. The lessons that can be learned based on this model can be used to identify the main reasons behind cart abandons and it may allow online retailers to work with these concerns in order to fix it and increase their conversion rates.

1.4 OBJECTIVES OF THE STUDY

The goals of this project are:

1. To locate and acquire worthwhile information about client behaviour, transaction history and web interactions data that influence cart abandonment.
2. To clean and analyze the data in order to identify the relationships and trends.
3. To develop predictive models on Random Forest algorithms.
4. To compare the models assessing them in terms of accuracy, precision, recall, and F1-score (Brownlee, 2021).
5. To make an interface that is easy to use so that the e-commerce managers can access the predictive system.
6. To provide recommendations in reducing cart abandonment on the basis of the result of the analysis.

1.5 METHODOLOGY

1. Data Collection: We are going to gather data on e-commerce websites, which contains data related to the customers, their preferences, purchasing history, and such data as how long they spend on the platform, how many pages they open, and click-through rate (Chen, 2022).

2. Data Preprocessing: We shall clean the data by filling the missing values, normalising the data and formatting the categorical variables.

3. Model Development: The random forest algorithm will be used to develop the predictive models since it works well when it comes to the classification and regression problems in the e-commerce analytics.

4. Model Evaluation: We shall evaluate the performance under accuracy, precision, recall and F1-score.

5. Interface Design: An interactive user interface will be created on the web-based platform by using the Streamlit model to engage users to read information and visualize future decisions.

1.6 SIGNIFICANCE OF THE STUDY

This paper will also assist the e-commerce companies learn more about reducing cart abandonment. With the help of the Random Forest algorithm, we will come to identify the trends in customer behaviour that will allow forecasting when it is most likely that a certain shopper will abandon his shopping cart. That knowledge can be used to make smarter decisions, convert more customers, and eventually, raise revenue (Chaffey, 2021).

1.7 SCOPE OF THE STUDY

This paper applies Random Forest algorithm to understand when customers would abandon their carts without completing their purchases to explore the issue of cart abandonment at e-commerce websites. This project covers data collection, preparation and cleaning, model construction and assessment; development of user interface that is very simple to enable the managers of e-commerce to derive actionable insights easily.

1.8 STRUCTURE OF THE PROJECT

This project will take this form:

Chapter One: Introduction

The chapter is the launching pad of the study as it gives a detailed background of the entire study that contextualizes the research undertaking. It will start with the development of the background of the researched phenomenon, presenting the current knowledge and motivation that led to conducting this research. After this, it is clear in stating the problem statement, which gives a clear definition of the gap in knowledge or the problem being resolved. The chapter further indicates the overall purpose of the study which is the general objective that the research seeks to attain which is in turn further divided into specific and measurable objectives that direct the process of carrying out the research. The methodology section presents the research approach in brief and techniques that will be adopted in the research with a view of offering clarity and rigor of the research process. Moreover, the chapter outlines the relevance of the study, which outlines its possible contributions as well as implications on practice. The project covered in the scope has also been neatly explained, which outlines the demarcations and constraints of the study. Lastly, an introduction gives direction to the reader by describing the flow of the following chapters where a preview of the content is provided.

Chapter Two: Literature Review

In the chapter, the researcher drills into the body of knowledge that is available in the field of inquiry to generate a critical analysis of the scholarly literature, industry reports, and other resources available in the field. Part one comprises the introduction and literature review of the e-commerce cart abandonment phenomenon to find out about its prevalence, reasons, and mitigation measures which are already in place. Later, the chapter explores on the theoretical background and the practical implications of predictive analytics in different areas with particular emphasis on how it is applied to tell the behaviour of the on-line consumer. Lastly, it gives a comprehensive overview of the machine learning methods that are core to this study especially with regards to the foundation, use and shortcomings of the Linear Regression and Random Forest algorithms in the field of predictive modelling. This extensive review builds the theoretical framework of this study and provides a gap in the given knowledge that the research will fulfil.

Chapter Three: System Analysis and design

In this chapter, a straightforward procedure to be followed in the analysis of these requirements and designing the proposed solution has been elaborated. System analysis stage includes the close study of current environment about e-commerce, and the issues on cart abandonment. This involves the identification of important stakeholders, the requirements of the users and establishment of the functional and non-functional requirements of the predictive system. After the analysis has been done, the system design process describes the solution blueprint. This considers the structure of the web application and dashboard, as well as the choice of right technologies of the frontend and backend development. It also reports the structure of a database to store relevant data, the user interface designing process including the application and the data visualization dashboard, and the system architecture to make the whole data processing flow, including data input, model training, prediction generation, and insight presentation. This chapter gives a better and explicit presentation of the way predictive system will be constituted and put into effect.

Chapter Four: Implementation and Results

In this chapter, the results of the study are provided clearly and in an orderly fashion whereby their results have been illustrated in a tabular or a figurative and other types of visual aids to effectively convey the results of the predictive modelling procedure. It thoroughly compares the quality of the built Linear Regression and Random Forest models according to the appropriate metrics, estimates the accuracy of their predictions, determines their advantages and drawbacks considering cart abandonment prediction. After giving the results, the chapter discusses the findings in detail. This implies the interpretation of the results regarding the research objectives and the literature reviewed in the previous chapters. The consequences of the model performance on e-commerce businesses are also discussed in detail, not neglecting the value of the predictive information obtained and possible ways to implement them in practice.

Chapter Five: Summary, Conclusion and Recommendations

The last chapter can be taken as the summary of the main findings of the whole research work. It gives a direct reiteration of the findings that are critical, and its statement of how far the research objectives have been achieved. The chapter is also critically reflective in that it summarizes the study in its limitations, that is, the limitations inherent in its study and how it might have worked. On the back of the conclusions that come out of the research, this chapter

provides practical solutions that e-commerce businesses can adopt to use predictive analytics to eliminate cart abandonment. In addition, it provides the possibility of future research directions, implying the topics where one might develop the findings of the present study and have more impact upon the comprehension of the intricate nature of online purchasing behaviour.

CHAPTER TWO

LITERATURE REVIEW

2.1 INTRODUCTION

This chapter gives an overview of the involved information that is critical regarding the project of using the Random Forest algorithm to build a prediction model of e-commerce cart abandonment analysis and forecasting. Intellectual sources it uses to draw its conclusions and ideas include but are not restricted to the academic journals, conference papers, scientific publications, and articles. Besides describing and analysing e-commerce websites and trends in customer behaviour, it also explains machine learning algorithms and specifically the ensemble methods and the history and criticisms of current and earlier cart abandonment prediction models.

The literature review explores the theoretical framework of consumer behaviour expressed in the online domain, discusses types of reasons why customers abdicate their online shopping carts and compares the predicting quality of distinct methods of machine learning applications in predicting customer behaviour. The ensemble methods and Random Forest algorithms are especially considered due to their proven success at being able to handle complex, high-dimensional data on e-commerce in such a way that such data remain somewhat interpretable by consumers of the business.

Moreover, another topic that is covered in the chapter is the problems of application of predictive models to real e-commerce systems such as data preprocessing methods, feature engineering problems, and model deployment challenges. The survey also speaks about ethics, such as using customer data and privacy concern within machine learning-based business intelligence systems.

According to the findings of this literature review, the effective model of Random Forests is developed to meet the practical needs of the contemporary e-commerce websites and effectively predict cart abandonment behaviour.

2.2 E-COMMERCE AND CART ABANDONMENT OVERVIEW

The world of online shopping is greatly different a decade ago and this has tremendously affected the way individuals utilize the websites to do their shopping. According to Statista (2024), the growth in e-commerce sales around the world has been pegged at 10.4 percent as compared to the previous year, with the 2023 sales amounting to \$6.2 trillion. This development has been accompanied with various challenges with the major one being the cart abandonment problem, which continues to haunt online businesses all over the world.

Cart abandonment is a phenomenon where customers add goods in their online carts and fail to make purchase. It is one of the greatest setbacks of e-commerce (Kukar-Kinney & Close, 2010). A comprehensive report by the Baymard Institute in 2024 shows that the global average of cart abandonment across sectors is 70.19 percent, which could get up to 85 percent in specific sectors. This reveals the relevance of knowing and forecasting the reasons that cause the abandonment of carts by customers.

The causes of cart abandonment derive out of a variety of famous models. Technology Acceptance Model In the Technology Acceptance Model developed by Davis in 1989, as also updated and expanded further by Venkatesh and others, 2003, the influence of the features of ease of use and perceived usefulness on online shopping behaviour is brought out in the following ways. Similarly, the Theory of Planned Behaviour that Ajzen created in 1991 describes the role of attitudes, social norms, and intentions in making decisions when making online purchases.

The research conducted by Kumar et.al (2023) published in the Journal of Retailing and Consumer Services was a follow-up study of this one. To facilitate online shopping they came up with the Digital Shopping Behaviour Model (DSBM). In their research they used 15,000 internet shoppers representing 12 different countries. They discovered that there is a combination of thoughts, emotions, and circumstances that count in the decision that people make to abandon their shopping carts.

2.2.1 Definition and Scope of Cart Abandonment

Cart abandonment is a major issue that significantly affects the e-commerce industry today. It happens when online shoppers put items in their virtual shopping carts but leave the website

without completing their purchase (Kukar-Kinney & Close, 2010). This simple act involves many consumer behaviors and decision-making processes, which collectively lead to billions of dollars in lost sales for online retailers each year.

Cart abandonment isn't just about whether a purchase is made or not; it includes various stages of the shopping journey. Research by Martinez et al. (2023) identified different points where cart abandonment occurs, from adding items to the cart to different steps in the checkout process. Their study, which analyzed 500,000 shopping sessions across 25 e-commerce sites, showed that 23% of abandonments happen within the first minute after adding items, 41% during information gathering, and 36% during checkout.

The way we measure and understand cart abandonment has changed as we learn more about online shopper behaviour. Previously, definitions only looked at cases where items were added to a cart without a purchase within 30 days. However, new research by Chen et al. (2023) suggests that we should consider other factors like cart value, session length, return visits, and purchasing behaviour over multiple sessions. Their long-term study of 100,000 customers over 18 months found that 34% of abandoned carts eventually resulted in purchases due to return visits or buying similar products.

The economic impact of cart abandonment is immense and continues to grow with global e-commerce expansion. The Baymard Institute's detailed 2024 study shows that the average cart abandonment rate across industries is around 70.19%, which translates to about \$4.6 trillion worth of abandoned goods each year. This figure becomes even more striking when considering global online retail sales reached \$6.2 trillion in 2023 (Statista, 2024), indicating that recovering abandoned carts could be one of the biggest opportunities for revenue in the digital market.

Different industries show varying rates of cart abandonment, providing insights into this phenomenon's nature and scope. Research by Rodriguez et al. (2022) examined patterns across 15 major industries and found significant differences: grocery had a rate of 56%, while fashion saw an abandonment rate of up to 89%. These variations are linked to factors like product type, average order value, how often people buy certain items, and how much research they do before buying. Industries with higher-value items that are purchased less

frequently tend to have higher abandonment rates due to perceived risks involved in making those purchases.

The timing aspect of cart abandonment adds further complexity to understanding it. Kim et al. (2023) studied time-related patterns and discovered that abandonments peak during lunch hours and late evenings; Tuesdays have the highest rates while Fridays have the lowest. Additionally, during busy shopping seasons such as Black Friday or holidays, cart abandonment rates can rise by an average of 15%, likely because shoppers tend to browse more and compare products during these times.

Researchers are also paying close attention to the psychological aspects behind why people abandon their carts. A significant study by Thompson et al. (2022) used brain imaging techniques during online shopping sessions and found distinct brain activity related to decisions about abandoning carts. Their findings indicate these decisions often involve complex interactions between thoughts about value and emotional responses influenced by social factors challenging traditional models focused mainly on price.

With mobile commerce growth and multi-channel shopping experiences becoming more common, technology's role in cart abandonment has expanded as well. Research by Wang et al. (2023) found mobile users abandon their carts at an average rate of 85.4%, compared to desktop users' rate of 67.8%. This difference arises from smaller screens making navigation harder and distractions inherent in using mobile devices. Additionally, customers often start shopping on one device but finish on another, complicating how we measure cart abandonment.

Cultural aspects are also becoming important as e-commerce grows globally and socially connected. Patel et al.'s research (2023) across 25 countries showed noticeable differences influenced by cultural values like individualism versus collectivism or uncertainty avoidance finding rates range from 52% in South Korea to as high as 89% in Brazil due to these cultural influences.

The rise of social commerce has introduced new dimensions affecting cart abandonment beyond conventional website interactions too. Liu et al.'s study (2022) found that social influences such as peer recommendations can significantly affect whether shoppers abandon

their carts; products with strong social proof had lower abandonment rates than similar ones without it.

As technology advances with tools like artificial intelligence and augmented reality changing how people shop online, our understanding of cart abandonment continues evolving too. Anderson et al.'s research (2023) explored voice commerce environments where visual cues are absent; they discovered unique patterns leading to different kinds of abandonments compared to traditional web experiences.

To truly grasp cart abandonment's full scope requires recognising its many dimensions, technical issues alongside psychological motivations, and social dynamics all interacting intricately together, influencing shopper behaviour today. Understanding this complexity helps create better strategies for reducing abandoned carts in modern e-commerce settings.

2.2.2 Factors Contributing to Cart Abandonment

The academic research has pinpointed many reasons why shoppers leave their carts without buying, which can be grouped into four main areas: technical, economic, psychological, and demographic factors. Knowing these reasons is vital for creating effective predictions about cart abandonment.

Technical Factors

Website performance plays a key role in whether customers complete their purchases. A significant study by Akamai Technologies (2023) showed that even a one-second delay in loading a page could lead to a 7% drop in sales. This was backed up by Chen et al. (2022), who looked at data from 500 e-commerce sites and found that pages taking longer than three seconds to load had 32% higher abandonment rates compared to faster ones.

Another major technical issue is the complicated checkout process. Research from Statista (2024) reported that 21% of people abandon their carts because the checkout takes too long or is too complex. This was confirmed by Rodriguez-Torrico et al. (2020), who analyzed 1,200 e-commerce websites and discovered that sites with more than four steps in the

checkout process saw abandonment rates that were 67% higher than those with simpler processes.

As mobile shopping grows, having websites that work well on mobile devices is more important than ever. A study by Wang et al. (2023) published in the *International Journal of Electronic Commerce* found that 78% of mobile users leave their carts when websites aren't designed for mobile use. Their findings stressed the need for responsive designs, easy-to-use interfaces, and payment systems suited for mobile users.

Economic Factors

Price sensitivity is a key reason behind cart abandonment. Research by Rajamma et al. (2022) in the *Journal of Business Research* studied pricing strategies on 200 e-commerce platforms and discovered that unexpected costs like shipping fees caused 55% of all cart abandonments. Their study showed that clear pricing could lower abandonment rates by as much as 40%.

Dynamic pricing adds another layer of complexity to how prices affect cart abandonment. A detailed study by Li et al. (2023) looked at how real-time price changes influence shopper behaviour and found that 34% of consumers abandon their carts when they notice price differences or frequent changes.

Preferences for currency and payment methods are also crucial factors. Research by Global Payments (2023) indicated that providing local payment options can cut cart abandonment rates by 25%. This finding is especially important for international online stores catering to various customers.

Psychological Factors

Researchers have recently focused more on the psychological aspects of cart abandonment. Decision anxiety, discussed by Sheth et al. (2022), impacts around 45% of online shoppers, particularly during expensive purchases. Their brain imaging study involving 200 participants revealed that the online shopping experience creates unique stress patterns leading to abandoned carts.

Trust issues and perceived risks remain major psychological hurdles as well. A meta-analysis conducted by Kim et al. (2023) reviewed 67 studies about online trust and concluded that

security concerns account for 19% of cart abandonments. The research emphasized the value of trust signals, security badges, and clear privacy policies in reducing these rates.

Social influences and peer pressure have become significant psychological factors today. Thompson et al. (2022) found that 28% of millennials abandon their carts due to distractions from social media or wanting social approval before making purchases. This highlights how closely linked online shopping is to social media platforms.

Demographic and Behavioural Patterns

Age plays an important role in cart abandonment habits. A longitudinal study by Davis et al. (2023) followed the shopping behaviours of 10,000 users across various age groups over two years and discovered that Generation Z (ages 18-25) has the highest abandonment rate at 78%, while Generation X (ages 42-57) has the lowest at 65%. The researchers attributed these differences to varying levels of comfort with technology and experience with online shopping.

Several studies have also documented gender differences in how people abandon carts. Martinez et al. (2022) found that women are 15% more likely to leave carts during product research, while men are 22% more likely to do so during checkout phases due to different shopping styles and decision-making approaches.

Geographic and cultural elements impact abandonment trends as well. A cross-cultural analysis by Patel et al. (2023) explored cart abandonment across 25 countries and noted substantial differences from a low of 52% in South Korea to a high of 89% in Brazil showing how essential it is for e-commerce designs and marketing strategies to adapt culturally.

2.3 MACHINE LEARNING AND PREDICTIVE ANALYTICS IN E-COMMERCE

2.3.1 Machine Learning Applications in Retail

The implementation of machine learning technology in e-commerce has transformed how retailers interact with and understand their customers. This use of artificial intelligence is

more than just a technological upgrade; it represents a significant shift in how businesses handle customer relationships, manage inventory, and generate revenue.

One of the most impactful applications of machine learning in e-commerce is customer segmentation. A study by Anderson et al. (2023) found that unsupervised learning methods, like k-means clustering and hierarchical clustering, can effectively identify different customer groups based on their shopping habits, browsing behaviour, and demographic data. Their research included 50,000 customers from three large e-commerce platforms and showed that using machine learning for segmentation made marketing campaigns 67% more effective compared to traditional demographic approaches.

Recommendation systems have become standard in e-commerce, with companies like Amazon, Netflix, and Spotify leading the way. A comprehensive review by Chen et al. (2022) examined the evolution of recommendation algorithms from collaborative filtering to those utilizing deep learning. Their results indicated that these advanced recommendation systems could increase average order values by 35% and boost customer retention rates by 28%. The study emphasized that hybrid methods combining content-based filtering with collaborative filtering are particularly successful.

Machine learning has also transformed demand forecasting by analyzing vast amounts of historical data along with seasonal trends and outside factors. Research by Kumar et al. (2023), published in the International Journal of Production Economics, showed that ensemble techniques like random forests and gradient boosting outperformed traditional time series methods when predicting product demand. Their case study involving a major online retailer demonstrated a 23% improvement in forecast accuracy, which resulted in a 15% drop in inventory costs.

Dynamic pricing is another area where machine learning has had a big impact. A study by Wilson et al. (2022) explored how reinforcement learning algorithms can adjust pricing strategies in real-time based on competitor prices, stock levels, and customer demand. Their findings across 12 product categories revealed an average revenue increase of 18% compared to fixed pricing models.

Additionally, fraud detection and prevention have significantly improved due to machine learning techniques. Research by Thompson et al. (2023), published in the Journal of Cybersecurity, found that ensemble methods can detect fraudulent transactions with an accuracy rate of 97.3% while keeping false positive rates low. Their work highlighted the importance of feature engineering in creating effective fraud detection systems.

2.3.2 Predictive Models For Cart Abandonment

The prediction of cart abandonment has changed from basic rule-based systems to advanced machine learning models. Knowing the different methods helps explain why Random Forest was chosen for this study.

2.3.2.1 Traditional Statistical Approaches

Logistic regression is the basic method used to predict when people will abandon their carts. It estimates the chances of cart abandonment by combining different features in a straightforward way, which makes it easy for business owners to understand. However, logistic regression has trouble with non-linear relationships and complicated interactions between features that are often found in e-commerce data. As a result, it only performs moderately well on actual datasets, with an AUC score of about 0.78.

2.3.2.2 Survival Analysis Techniques

Survival Analysis looks at cart abandonment as a time-to-event issue, focusing on when the abandonment happens instead of just whether it happens. The Cox Proportional Hazards Model and Kaplan-Meier estimators can accurately reflect changes over time. However, these techniques depend on certain assumptions about hazard rates and survival distributions that might not apply to every e-commerce situation, which can restrict how widely they can be used.

2.3.2.3 Support Vector Machines (SVM)

SVM is great at working with high-dimensional data by using kernel functions that can identify non-linear connections. The RBF kernel SVM does a good job with complicated datasets, achieving an F1-score of about 0.85. However, SVM has some downsides: it can be

very slow with large datasets, it needs features to be scaled properly, it requires careful adjustment of its settings, and it's hard to understand how it makes decisions, which is important for business use.

2.3.2.4 Neural Networks and Deep Learning

Neural networks can learn complex patterns from raw data on their own, without needing manual feature engineering. Deep feedforward networks can reach high accuracy (around 87.4%) when working with large datasets. However, they need a lot of training data, can easily overfit the data, are often seen as "black boxes" because they are hard to understand, and require a lot of computing power and careful adjustment of settings.

2.3.2.5 Long Short-Term Memory (LSTM) Networks

LSTM Networks are great at understanding patterns in how users behave over time, especially when looking at their clickstream data. They perform 15% better than older methods when predicting events over time. However, LSTMs need the data to be organized in a certain way before they can work, use a lot of computer power, are hard to understand, and have trouble handling sequences that vary in length, which often happens in e-commerce sessions.

2.3.2.6 Decision Trees

Decision Trees are easy to understand and create clear business rules that everyone can follow. They work well with different types of data and don't need much preparation. However, individual decision trees can easily overfit the data, meaning they might not perform well on new data. They can also be unstable, as small changes in the data can lead to big differences in the tree structure. Additionally, their ability to predict outcomes is not as strong as that of ensemble methods.

2.3.2.7 Naive Bayes Classifiers

Naive Bayes is quick to compute and is effective with categorical data, which makes it a good choice for real-time uses. It does a decent job overall (AUC ~0.74) even though it is

simple. However, it assumes that features are independent from each other, which is often not true in actual e-commerce data. This can reduce its ability to make accurate predictions when the features are related.

2.3.2.8 Why Random Forest Was Selected

Due to the weaknesses of individual algorithms, this study selected Random Forest because it solves most of these issues:

Benefits compared to other methods:

- It manages non-linear relationships better than logistic regression.
- It's easier to understand than SVM and neural networks, thanks to feature importance.
- It is less likely to overfit than single decision trees.
- It is more efficient in terms of computation when compared to deep learning methods.
- It can naturally work with different types of data without needing extra preparation.
- It offers estimates of uncertainty through its voting system.
- It needs less adjustment of hyperparameters compared to SVM and neural networks.
- It works well with large e-commerce datasets.
- It has built-in feature selection by randomly sampling features.

When it comes to predicting cart abandonment, this approach:

- Understands the complicated ways user behaviour interacts.
- Deals with missing data easily, which is often found in e-commerce.
- Offers rankings of important features for better business understanding.
- Strikes a balance between being accurate and easy to understand.
- Works effectively with unbalanced datasets, which are common in cart abandonment scenarios.

Is ready for real-time use with quick prediction times.

Random Forest blends the clarity of decision trees with the strong prediction abilities of ensemble methods. This makes it perfect for business situations where both accuracy and clarity are important for systems that predict cart abandonment.

2.3.3 Random Forest Algorithm in Business Context

Random Forest algorithm has emerged as one of the most flexible and successful methods of machine learning in business applications and particularly forecasting cart abandonment. Its success rate, effectiveness, readability, and dependability have caused it to emerge as a preferential choice of many of the experts and scientists in the field of e commerce.

Random Forest owes its performance to an ensemble-based learning where a number of weak learners form the strong one. Random Forest was proposed by Breiman (2001) as a variation of bagging that is based on bootstrap aggregating and random variables. This is a beneficial combination of randomness that has a positive effect of reducing overfitting and improving the generalization performance when the model reliability is key especially in a business setting.

The evaluation of the Random Forest based on the adequate use of Random Forest in each of 20 business prediction tasks like customer churn, fraud detection, and demand forecasting evaluated an extensive study by Kumar et al. (2023). Their results always ranked Random Forest in the top three algorithm as it performs especially well in handling mixed data types of data and lack of values often involved in real-world business data sets.

The capability of Random Forest to identify the importance of features has assisted in the interpretation of business requirements immensely. Anderson et al. (2022) research indicated that, in this way, these importance scores can provide useful information to business leaders. When they did the study on cart abandonment prediction they identified product price, time spent on product pages and checkout steps to facilitate the prediction. This expertise resulted in specific measures which reduced rates of abandonment by 28%.

Random Forest is also attractive because of its ability to handle missing data without involving imputation which is very attractive in terms of business applications. Chen et al. (2023) compared several strategies of missing data processing in terms of several algorithms and identified that inbuilt strategy of handling missing data in Random Forest performed as well as more sophisticated approaches but required significantly less preprocessing.

It is due to the different methods that help to enhance the clarity of Random Forest models. Wilson et al. (2022) explored the possibilities of using SHAP (SHapley Additive exPlanations) values and Random Forest models to provide interpretations of each prediction.

The technology they created on a large e-commerce platform allowed customer service reps to know why some specific customers were likely to leave their carts so they could make more of a direct effort to retain them.

Computational efficiency is the other advantage of Random Forest in the business context. Thompson et al. (2023) conducted a benchmarking study, where several algorithms with real-life business data sets were put to comparison in terms of training and prediction times. They have discovered that Random Forest scaled to dataset size and their training times were linear but their low prediction times were suitable in real-time applications.

Lastly, various research revealed Random Forest as being resistant to alteration of hyperparameters. According to research by Rodriguez et al. (2022), Random Forest maintains consistent performance over a broad range of hyperparameter settings and, therefore, it is rather suitable in the business context, where hyperparameter optimization may not be feasible.

2.4 DATA ENGINEERING AND IMPLEMENTATION FRAMEWORK

2.3.1 Data Sources and Feature Engineering

How well machine learning models can predict the cart abandonment significantly depends on the data whose quality and relevance are relied upon during the training process. E-commerce sites capture a lot of information in various forms and hence it is crucial to understand how to tap this information to build powerful models of predictive application.

Information content on web analytics is the major source of information used to forecast cart abandonment. By analyzing 50 large e-commerce websites, Kim et al. (2023) identified more than 200 possible features that may be extracted based on the user browsing behaviour. According to their study, variables such as the time browsing a page, the extent to which a user scrolls and mouse movement patterns were particularly effective in predicting abandonment behaviour.

Customer demographic information can be used to offer a background of why a cart is abandoned. Davis et al. (2022) conducted in-depth research into the implications that demographic variables can have in predicting behaviours in various customer groups. They

found age, income and location to be important predictors, but they mattered very much based on product types and price points.

Information regarding product catalogue and pricing provides information on the dependence between product characteristics and behaviour of abandonments. In studies, Martinez et al. (2023) investigated the role of the price, category, brand, and customer reviews that can influence cart abandonment decisions. In their examination of 2 million product interaction information, products that received high review scores and those that have great product details experienced 35 percent fewer product abandonment incidences.

History of transactions can provide important insights on customer behaviour and preferences. According to the study published by Patel et al. (2022) that followed the previous buying behaviour of 100,000 clients throughout three years showed that the past behaviour foretells future abandonment trends forcefully. Customers who had already made purchases were 45 percent less prone to leave the cart in the future.

External databases have begun to gain some interest since they can increase prediction quality. Chen et al. (2023) used weather, economic, and sentiments of social media in their cart abandonment models. Their results revealed minor yet significant gains in the accuracy of the prediction, particularly on products that have seasons or weather dependence.

The process feature engineering is crucial in the transformation of raw data to predictor. The study by Liu et al. (2022) analyzes many diverse approaches to feature engineering of e-commerce data. In their study, they identified some of the most important categories of engineered features that kept increasing model performance.

Behavioural characteristics will show the level of interaction of users with the site. Anderson et al. (2023) developed an extensive model of how to derive behaviour features given a clickstream data set. They incorporated such metrics as the number of times one visits a page, the duration of the visit, and the intensity of interaction and all these contributed to the improvement of their models by 18%.

Temporal features This has to do with patterns of time that are relevant in conceptualising abandonment behaviour. This finding demonstrated by Wilson et al. (2022) indicated that

time of day, day of the week, and time since last visit could all be effective factors increasing prediction accuracy by a significant margin. Their efforts in temporal feature engineering have resulted in the 23 percent increase in the F1-score of cart abandonment prediction.

Aggregation features summarise the statistics that reflect customer behaviour with time. Brown et al. (2023) examined several results obtained by aggregation, specifically using rolling averages and calculating trends, seasonal disaggregation. They concluded that features involved in aggregation were most helpful in terms of detecting long-term trends in behaviour.

Text mining functionality translates useful information about product descriptions, reviews and communication between the customers. The work by Zhang et al. (2022) is an application of natural language processing methods of detecting sentiment and topic on product reviews. This solution enhanced the prediction by 12 percent in those products that have sufficient review data.

2.4.2 Model Implementation and Deployment

This transition between model development and implementation in production is a milestone in any machine learning project, and model-based prediction of cart abandonment is important to make swift decisions. The implementation approach must strike the balance among precision, time, and grow capabilities and the capability to exhibit reliable performance in the practical world.

Also, the decision to split predictions to either a real-time or a batch is a simple architecture decision. Findings of a study conducted by Kumar et al. (2023) show the results of the real-time and batch prediction systems at the 15 e-commerce platforms and their resource requirements. They found that real-time systems were 34 percent more likely to be able to act because of their timely predictions but they needed thrice as much processing power and more complex configurations.

Work with the constant streams of statistical data of user behaviour is getting more demands of stream processing frameworks. Thompson et al. (2022) provided clear exploration of the various stream processing systems such as Apache Kafka, Apache Storm, and Apache Flink

to predict cart abandonment. According to their results, Kafka Streams was the fastest, with low latency and easy to use in most cases in the e-commerce space.

The system of serving models has to handle very high concurrent requests and with minimal delays. According to the research conducted by Rodriguez et al. (2023), a wide range of model serving platforms was tested including TensorFlow Serving, MLflow, and custom REST APIs. They found specialized model serving platforms provide more reliable production systems, because of the improved performance monitoring and management of models.

The feature stores are necessary to manage and provide features in development environments. Chen et al. (2022) conducted a study that demonstrated that feature store could help in the consistency of models, without the need to invest much effort in feature engineering. With their methodology they reduced the time it takes to deploy their models by 40 percent and increased the use of features in other modeling work.

The A/B testing frameworks play the crucial role of validating the success of models in a production environment. A study by Davis et al. (2023) developed an A/B testing model, which is applicable to the prediction models of cart abandonment in particular. Their algorithm involved a statistical power analysis, experiment design and automation of result reviews, which reinforced the model validation.

Development of various versions of models is important in the production where frequent updates occur. The effect that several methods to version models have on the reliability of the system was examined by Wilson et al. (2022). In their findings, they emphasized on the role of having automated rollback and incremental deployment.

Finally, monitoring and alerting tools are required to ensure that model performance can be tracked even when it is in production. Patel et al. (2023) engaged in research by creating a universal observational framework that monitors the accuracy of the model, the alteration of data, and general performance. With this approach, it was possible to anticipate any degradation of model in time and system-reduce power to 60%.

2.4.3 Business Impact and Intervention Strategies

The possible use of cart abandonment prediction models to improve business performance through strategic action provides the real capacity of this type of models. We ultimately need to learn how to turn the predictions into action, i.e., how to make forecasted initiatives as effective as possible in terms of investment returns that we may make with such initiatives.

Email marketing can be classified as one of the most used techniques in a cart abandonment strategy. In a study by Martinez et al. (2023), the article tested the effectiveness of the multiple approaches to email across 200 e-commerce sites. They have found that the personalized email campaigns which were activated due to the prediction of abandonment have demonstrated a conversion rate, which was 25 percent higher in comparison to the generic email.

Retargeting ads also tend to help in rescheduling customers who have abandoned items on their carts. The authors of the study performed a vast amount of research and already tested a variety of retargeting approaches to various segments of customers (Liu et al., 2022). After trying out the dynamic retargeting ads based on the abandoned products, they found that the click-through rates were about 35 percent more and their conversion rates augmented by 28 percent as compared to the static advertisement.

Making offers in terms of individual lowered prices may be the valuable approach to transforming abandoned carts into sales. Anderson et al. (2023) researched how machine learning models could aid in the improvement of discount strategies with reference to terms of customer behaviour and chances of abandonment. Their strategy using dynamic prices led to them increasing the conversion rates by an outstanding 42 percent, and they were also able to extend their profit margins.

With the mobile commerce, intervention opportunities can be conducted within seconds through push notifications and alerts. Kim et al. (2022) used the data to recover the abandoned carts with regular push notifications to test their influence on 50 applications in mobile commerce. Their findings showed that timely and personalised notifications had the potential of recovering up to 18 percent of abandoned carts.

By using social proof and sense of urgency it is possible to stimulate the customers into making their purchases because of appealing to their psychology. Brown et al. (2023) conducted a study in which they tried to minimize cart abandonment by testing several psychological triggers on 30 online shops with the help of A/B testing. Their results revealed that the disclosure of stock balance and past purchases of other customers resulted in a decrease of the abandonment rates by 22 percent.

Preventive interventions are aimed to prevent the abandonment and reduce it in advance. The framework proposed by Zhang et al. (2022) has the benefit of real-time predictions that allow starting interventions at the shopping sessions, which allows reducing the abandonment rate by 31% in comparison to common reactive approaches.

Real-time personalization is one of the new practices that will personalize shopping experience according to the probability of abandoned carts. Thompson et al. (2023) performed dynamic personalisation on their websites, refashioning content, graphic design, and messages in accordance with the real-time forecasts. The novel system realized a 26% decrease in the abandonment rates and an 18 percent increase in average order value.

Besides, the customer service tools may have a notable effect on high-value potential customers, abandoning their cart. Philips et al. (2019) examined the use of live chat and proactive customer service contact that may be activated based on abandonment predictions and found that the former intervention converted 45% of high value abandoned carts.

Lastly, the calculation of ROI and business value is essential when it comes to justification of such investments into machine learning technologies. The framework designed by Rodriguez, et al., (2023) presents a general overview of the application of predictive models of the cart abandonment business, considering the direct effect on income, the change in customer lifetime value, and the effect on operational costs.

2.5 RESEARCH GAPS AND THEORETICAL FRAMEWORK

2.5.1 Literature Gaps and Limitations

Despite the numerous studies that exist on how to predict cart abandonment and how machine learning can be applied in e-commerce current literature has a considerable amount of gaps. These gaps therefore provide worthwhile opportunities to further research on and provide stronger methods to comprehend and predict cart abandonment behaviour.

There is one major gap which is concerned with lack of holistic research on real-time implementation. Although most studies demonstrate the theoretical success of different machine learning techniques, almost no study examines into the production hurdles and performance results when the techniques are used in production settings. The study by Kumar et al. (2023) found that only a quarter of published works disclosed information regarding actual implementation in real-time and even less reported long-term performance of systems in practice.

The other identified limitation is the poor comparative study of ensemble techniques employed in the prediction of cart abandonment. Despite there being individual studies on the content of such techniques as Random Forest and Gradient Boosting, it is difficult to find comprehensive comparisons based on standardised datasets and evaluation criteria. Such a drawback makes the selection of the most appropriate algorithm to be used by the practitioners more complicated.

There are also no such strong frameworks of the feature engineering in existing literature focused on e-commerce data. Although the types of features have been studied in a variety of cases, there is not a common approach to feature engineering to predict cart abandonment. According to the study by Anderson et al. (2022), more than 50 unique feature engineering methods were identified in the variety of studies, but an absence of commonality was found in the techniques or measures.

Also, the cross-cultural and geographic differences in the cart abandonment behaviour are not adequately researched. Most of the studies focused on North American and European markets, and the existing knowledge gap includes the way people abandon in an emerging market or in different cultural conditions. Such a drawback decreases the lavishness of discoveries and limits the application of models to other global markets.

Besides, the temporary nature of cart abandonment behaviour has been largely underestimated in research discourses. Even though other scholars have incorporated the element of time, not many have explored the aspect of how the patterns of abandonment change over time or how a model can be modified as the behaviour of the customers change. This supervision is highly topical since the world of e-commerce was changing rapidly and the trends of consumers were constantly changing.

Finally, there is little knowledge concerning the coupling of predictive models and business processes, including decision-making workflow. Most studies pay more attention to accuracy and technical performance of the models and fail to examine how these models can be integrated into the processes and decision-making frameworks of the companies without any inconveniences.

2.5.2 Ethical Considerations and Privacy

The increasing adoption of machine learning in forecasting cart abandonment raises both critical concerns of ethical matters and privacy concerns that require caution in use to ensure their responsible application. Such an aggregation and use of detailed behaviour data to make predictions induce significant privacy, consent and fairness in algorithms challenges.

The privacy of data, particularly the General Data Protection Regulation (GDPR) introduced in the European Union and other regulations across the world, influence the systems of predicting cart abandonment greatly. A study conducted by Thompson et al. (2023) studied the challenges e-commerce sites should undertake when making such predictions with machine learning. They found out that 67 percent platforms could not get effective consent of collecting behavioural data and 45 percent of platforms faced difficulties in adhering to regulations concerning deletion of data.

There is not much literature in the previous studies related to the problem of algorithmic bias, which is predictive of cart abandonment. In a study carried out by Davis et al. (2022), it was observed that the cart abandonment models might include biases such as demographic bias, behavioural bias, and temporal bias. They found out that models constructed on historical data may strengthen the historically existing biases and treat some groups of customers unfairly.

The ethical questions or the need to have a balance between the business interests and the ethical questions make fairness in predicting cart abandonment a complex issue. A study by Martinez et al (2023) designed an evaluation framework to test fairness in machine learning systems in e-commerce, such as parity demographic, equalised odds, and individual fairness such evaluation platforms. In their work, they indicated the trade-offs between the accuracy of the predictions, as well as fairness across different group of customers.

Trusting customers and complying with the law are increasingly becoming dependent on transparency and explainability. According to Liu et al. (2022), they have empirically examined how cart abandonment prediction can be made easier to explain, possible solutions include LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations). In their studies, it could be seen that explainable models could still be good in making their predictions and explain exactly how it happened.

The consent and control of information possessed by the client is the issue that arises as long as system applied to make a prediction of carts that have left the limitation. Investigations conducted by Chen et al. (2023) examined the customer attitudes towards sharing behaviour data and concluded that 78 percent of all customers would be eager to provide their data to receive a specific advantage in exchange, such as personalised suggestions or special advertisements.

2.5.3 Conceptual Framework For Current Study

Having thoroughly analysed the current studies on the subject, this research paper suggests a conceptual framework which combines the key discoveries and fills the informational gaps that have been created in the field. This framework is aimed at helping formulate a

framework on Random Forest that can simply predict the cart abandonment and become easy to work within its implementation.

The framework is established on several popular theories, which are the Technology Acceptance Model (TAM), the Theory of Planned Behaviour, and the Digital Shopping Behaviour Model. The theories will give an insight into the factors that affect online shopping behaviour and reasons as to why buyers end up leaving their shopping carts.

The framework consists of four primary blocks, such as data collection and preprocessing, feature engineering and selection, model development and validation, and implementation and evaluation. Each element responds to a certain lacking in the research identified in the literature review.

The data collection section is where we stress the necessity to collect comprehensive data and represent the information of the various sources variously web analytics, customer demographics, product information and transaction history. The ethical issues along with privacy laws are also considered at the very beginning of the framework so that no laws were violated and the expectations of customers were met.

The feature engineering section shows how to organize the process of developing predictive features using simple data of the e-commerce. It embraces characteristics of behaviour, those of time, those of aggregation, and those of context and centres on what Random Forest algorithms require.

The model development segment is to enhance the performance of Random Forest to ensure that businesses can understand it. The framework also has comprehensive measures of evaluation such as cross-validation measures, performance measures and business impact evaluation measures.

In the implementation section the authors look at factors on the ground associated with the practical difficulties in implementing machine learning models. It considers such things as real-time predictions, system integration, watch as well as constant maintenance.

The research questions to the study are:

1. What could be done to optimize Random Forest algorithms of predicting cart abandonment and online shopping?
2. Which methods of feature engineering can be used to improve the accuracy of predictions?
3. What are the ways to overcome real-time implementation issues without affecting model performance negatively?
4. How do cart abandonment prediction using machine learning systems affect businesses?

This research study is meant to contribute:

- An elaborate feature engineering framework of the prediction of cart abandonment
- Better utilization of Random Forest in web-based retailing
- Verification on real-life of model performance in real worlds
- Down-to-earth recommendations on building cart abandonment forecasting mechanisms
- Cost and benefit measurement of business and ROI

2.6 CHAPTER SUMMARY

This literature researched recent articles on cart abandonment prediction and machine learning as it relates to e-commerce. The key points are the following:

Impact of the Problem: The situation of cart abandonment occurs in more than 70 percent of online shopping experiences, and this is a major issue that companies require substantial solutions to predict effectively.

Algorithm comparison: Various machine learning techniques are promising but Random Forest was selected because it has a very good balance of value (accuracy), tractability and ease of use relative to other techniques such as SVM, neural networks, or individual decision trees.

Research Gaps: The available literature demonstrates the absence of the researches on a real-time implementation, a limited literature on comparison of ensemble methods, and insufficient studies on the contribution of those issues to the businesses and challenges in implementing them.

Ethical Issues: Special attention should be paid to the aspect of privacy, approval, and equity in introducing such algorithms.

This review not only preconditions the development of the cart abandonment prediction mechanism based on Random Forest that can cover the gap detected in the literature and can be of use to both academicians and actors of the e-commerce world. The following chapter is going to clarify how one can translate this theoretical base into a practical research plan.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

3.1 INTRODUCTION

This Systems Analysis part of the chapter gives a summary of the entire system and closely examines both the functional and non-functional requirements. The design section includes the architectural design of the system and uses different diagrams to show how the system works.

3.2 SYSTEM ANALYSIS

System analysis is a methodological approach to comprehending and documenting the requirements and functionalities of a system. It involves collecting and analyzing information to identify the system's needs and goals. By examining existing documentation, studying system operations, and analyzing data, system analysts break down the system into its constituent parts and examine their interrelationships. The primary objective is to determine the most effective solution to meet these needs and enhance system efficiency. This phase generates documentation, specifications, and models that provide guidance for the subsequent design and development of the system. System analysis is essential for ensuring that the system aligns with its objectives and enhances its performance

3.3 ANALYSIS OF EXISTING MODELS

Existing methods of cart abandonment encompass a range of approaches, such as logistic regression and decision trees to analyze abandonment patterns (Smith et al., 2016). More advanced implementations employ neural networks to capture non-linear use behaviour, though these often lack interpretability (Zhang & Lee, 2018). The Online Shoppers Purchasing Intention Dataset (UCI, 2018) has become a benchmark for training these models, providing session-level data on 12,330 user interactions. Each method has its own strengths and considerations. Logistic regression in cart abandonment prediction excels at identifying which specific factors (page load time, number of items, price thresholds) directly influence customers leaving their carts, making it easy for e-commerce teams to understand and act on insights like "customers abandon 40% more when shipping costs exceed \$15", but struggles

to capture complex shopping behaviours like browsing patterns combined with seasonal trends or the interaction between product categories and user demographics. Decision trees provide clear, actionable rules for cart abandonment, like "if cart value > \$200 AND shipping > \$20 THEN 85% abandonment probability," making them perfect for creating targeted intervention strategies and explaining decisions to marketing teams, though they often oversimplify complex customer journeys and may miss subtle behavioural patterns like gradual price sensitivity changes across different customer segments.

Neural networks excel at detecting intricate cart abandonment patterns, such as identifying customers who browse extensively before abandoning based on their click sequences, session duration, and product interaction patterns and capturing non-linear relationships between factors like time-of-day shopping and abandonment rates, but their black-box nature makes it impossible to explain to business stakeholders why a specific customer was flagged for intervention or what specific changes would reduce abandonment.

Ensemble Methods effectively combine multiple signals of cart abandonment risk such as browsing behaviour, product preferences, and session characteristics, while providing feature importance rankings that help e-commerce teams prioritize which factors to optimize (like checkout flow vs. shipping options), though they require more computational resources for real-time abandonment predictions during active shopping sessions.

Deep learning approaches can model complex sequential shopping behaviours like identifying customers who typically abandon after comparing products across multiple sessions, learning from entire customer journey histories including past purchases and returns, and detecting subtle patterns in how customers interact with product images and descriptions, but their complexity makes them unsuitable for explaining abandonment decisions to customers or creating transparent intervention strategies.

Hybrid approaches allow e-commerce platforms to use interpretable models to understand why customers abandon (high shipping costs, complex checkout) while employing complex models to predict which specific customers will abandon in real-time, enabling both strategic business insights and tactical intervention campaigns, though they require sophisticated infrastructure to manage multiple prediction systems and potential conflicts between different model recommendations. When selecting a method, it is important to consider the specific

characteristics of the data and the desired balance between interpretability and predictive accuracy

3.3.1 Drawbacks of Existing Systems

Current e-commerce systems designed to address cart abandonment include logistic regression models for basic predictions, decision tree algorithms for rule-based actions, neural networks for recognizing patterns, ensemble methods for better accuracy, and hybrid frameworks that mix different techniques. However, these systems have several important limitations to consider when assessing their performance:

1. **Limited Real-Time Intervention:** Most cart abandonment systems work by processing data in batches and analyzing past behaviours instead of detecting issues in real time. This means they can only send recovery emails hours or days after a customer leaves their cart, rather than stepping in while the customer is still shopping and more likely to make a purchase.
2. **Generic Intervention Strategies:** Many existing systems use a one-size-fits-all strategy, sending out generic discount emails or standard remarketing ads. They do not customize their approaches based on different customer groups or reasons for abandoning carts. For example, they might not offer fast shipping to customers who need items quickly or discounts to those looking to save money.
3. **Lack of Multi-Channel Data Integration:** Current systems often look at cart abandonment separately from other customer interactions. They usually do not include data from social media, email activity, mobile app usage, or visits to physical stores. This leads to an incomplete picture of customer behaviour and missed chances for recovery strategies across different channels.
4. **Poor Insights and Interpretability:** While many advanced machine learning models are accurate in predicting cart abandonment, they often do not explain why specific customers leave their carts or suggest what actions could reduce these rates. This makes it hard for e-commerce teams to make systematic improvements to checkout processes or product offerings.

5. Challenges with Dynamic Customer Context: Current systems have difficulty considering real-time factors like inventory levels, promotional events, competitor prices, seasonal demand changes, and individual customer situations (like budget limits or urgency). As a result, their interventions may be poorly timed or irrelevant, which can hurt customer relationships instead of helping recover lost sales.

3.4 ANALYSIS OF PROPOSED SYSTEM

The proposed cart abandonment prediction system will be built using a random forest classifier with 100 decision trees, a maximum depth of 15, and a minimum sample split of 10, configured within a scikit-learn preprocessing pipeline that handles both numerical features (administrative pages, durations, bounce rates, and exit rates) and categorical features (month, visitor type, browser, and region) through standardization and one-hot encoding. The system will have multiple interfaces accessible to e-commerce business analysts, marketing teams, and store managers who can predict customer abandonment likelihood, visualize session behaviour patterns and feature importance through interactive dashboards, and implement targeted mitigation strategies. It will also include a Django-based web application for individual customer predictions, a Streamlit dashboard for batch analysis and real-time visualization of abandonment trends, and a REST API for seamless integration with existing e-commerce platforms, enabling users to input customer session data and receive immediate risk assessments along with automated recommendations such as discount codes for high-risk customers or cross-selling opportunities for low-risk customers.

3.4.1 Benefits of Proposed System

1. Proactive Revenue Recovery Through Early Intervention: The system can predict when customers are likely to abandon their carts in real-time, which lets businesses step in before they leave. Instead of waiting for customers to receive a post-abandonment email, companies can act quickly by recognizing risky sessions as they happen. They can then offer discounts or live chat support right away, helping recover sales that might have been lost. This proactive strategy can greatly enhance conversion rates and lower the usual 70% cart abandonment rate seen in e-commerce.

2. **Enhanced Customer Experience Through Personalized Interactions:** By analyzing which factors most affect customer choices, the random forest model helps create personalized responses. For example, if it notices a customer is sensitive to prices, it can send a discount offer automatically. On the other hand, if a customer shows interest in products, it might suggest related items for purchase. This level of personalization makes shopping more relevant and enjoyable, fostering customer loyalty.

3. **Operational Efficiency Across Multiple Business Functions:** The multi-interface design meets different organizational needs at once. Business analysts can dive deep into data using the Streamlit dashboard, marketing teams can track real-time trends and how well campaigns are performing, and store managers can monitor individual customer interactions through the Django web application. The REST API integration ensures everything runs smoothly within current workflows, removing the need for manual data exports or separate analysis tools.

4. **Data-Driven Decision Making with Actionable Insights:** The system changes raw session data into clear insights by examining patterns like which administrative pages were visited, how long sessions lasted, bounce rates, and exit rates alongside factors such as visitor type and browser used. The interactive dashboards visually show customer behaviour trends and feature importance, helping teams make smart decisions about improving websites, planning marketing strategies, and managing inventory based on solid evidence rather than guesses.

5. **Scalable Automation with Intelligent Resource Allocation:** The automated recommendation engine enhances marketing efforts by directing resources to the best opportunities. Instead of sending out blanket discount offers to everyone, the system wisely targets promotions only at high-risk customers who are likely to respond well while also identifying low-risk customers for upselling chances. This focused approach maximizes returns on marketing investments and cuts down unnecessary discounts that could hurt profit margins.

3.5 REQUIREMENTS SPECIFICATION

Before designing and implementing a system, it is essential to specify detailed requirements to ensure effective and efficient analysis and design. The requirement specification serves as

a description of the software system to be developed, outlining its features and behaviours. It consists of functional and non-functional aspects that define the system's functionalities and qualities. The following are the functional requirements of the system:

3.5.1 Functional Requirements

Functional requirements outline the essential abilities and specific tasks that the cart abandonment prediction system must carry out to achieve business goals. These requirements clarify what features and functions users can anticipate from the system, ensuring it provides measurable benefits in predicting and reducing instances of cart abandonment (Saleh,2022). The following are the functional requirements of the system:

Data Management: The system needs to upload and prepare e-commerce transaction data by importing and cleaning raw information from various sources for analysis. The feature engineering pipeline converts this raw data into 17 important features, such as pages visited, time spent on each page, and bounce rates, which are used for training the model.

Machine Learning Operations: The system is required to train a Random Forest Classifier with 100 trees, a maximum depth of 15, and a minimum sample split of 10 to predict abandonment. It will generate real-time predictions by delivering instant cart abandonment risk scores for users currently active on the site through a web interface and API.

Data Storage: The system will save prediction results in a Django database to keep track of prediction outcomes, model performance metrics, and user session data for future analysis. Model serialization using joblib enables saving and loading trained models to ensure consistent predictions even after system restarts.

Visualization and Reporting: The web dashboard created with Django and Streamlit will display interactive interfaces for viewing individual predictions, processing batches, and visualizing feature importance. Tracking performance metrics will continuously monitor accuracy, precision, recall, F1-score, and AUC-ROC scores.

Intervention System: The automated recommendation engine will initiate targeted actions like discounts and popups when high risks of abandonment are predicted. API integration will

allow for REST endpoints that enable seamless connection with existing e-commerce platforms.

3.5.2 Non-Functional Requirements

The proposed system has several non-functional requirements:

Scalability: This means the system can efficiently manage growing amounts of data and more users at the same time without slowing down or crashing.

Performance: This includes how quickly the system can generate predictions and load dashboards, aiming for real-time responses within 2-3 seconds for the best user experience.

Accuracy: This measures how correct the model's predictions are, using sklearn's `accuracy_score` function to ensure it reliably predicts cart abandonment while being consistently monitored for performance.

Usability: This focuses on making web interfaces simple and easy to use, requiring minimal training so that business users can navigate and make use of the system features effectively.

Security: This involves protecting customer information through GDPR compliance and secure API authentication methods to block unauthorized access.

Reliability: This ensures the model performs consistently by using a random state seed of 42 and ongoing performance checks to provide stable and trustworthy results over time.

Maintainability: This refers to the system's modular design, which separates the Django backend, Streamlit frontend, and API layers. This setup makes it easier to update, debug, and maintain the system.

3.6 SYSTEM DESIGN

System design is the process of defining the overall structure, components, modules, interfaces, and data for a system to satisfy specified requirements. It focuses on how the

system will solve a problem, including the organization of its parts and their interactions. System design translates functional and non-functional requirements into a blueprint for building the system.

3.6.1 Architectural Design

Architectural design involves comprehending the structure of a system and creating a cohesive framework for its components. It demonstrates how these components collaborate to fulfill the system's main objectives. Architectural design focuses on identifying the subsystems comprising the system and establishing the means for controlling and facilitating communication among these subsystems. It encompasses the overall organization and arrangement of the system's elements, ensuring effective coordination and interaction between them.

The cart abandonment prediction system follows a multi-layered architecture that processes e-commerce transaction data through distinct stages to deliver accurate abandonment predictions. The system architecture comprises six primary layers, each serving a specific function in the prediction pipeline.

Presentation Layer: This is the user interface layer where business users interact with the system through web-based dashboards. Users can access the system via Django web application or Streamlit interface to input customer behavioural data, view real-time predictions, and access historical analytics. This layer provides dropdown menus for selecting visitor types, input fields for behavioural metrics, and visualization components for displaying prediction results and feature importance.

Data Capture Layer: This layer handles the ingestion of e-commerce transaction data from multiple sources, including CSV files, JSON formats, and database connections. The system captures real-time customer behavioural data such as page visits, session durations, bounce rates, and exit rates. It will also collect categorical information, including visitor types, operating systems, browsers, regions, and traffic sources, establishing the foundation for prediction analysis.

Data Processing and Cleansing Layer: After data capture, this layer will perform comprehensive preprocessing operations to ensure data quality and consistency. It handles missing values through imputation techniques, scales numerical features using

StandardScaler to normalize data ranges, and encodes categorical variables using OneHotEncoder. The layer removes unwanted columns, eliminates zero values where inappropriate, and transforms raw behavioural data into the 17 key features required for model training.

Feature Engineering Layer: This layer transforms the cleaned data into meaningful input features for the machine learning model. It creates derived features from raw behavioural data, applies feature scaling and normalization, and structures the data into the format expected by the Random Forest classifier. The layer ensures that all 17 features, including administrative pages, durations, bounce rates, and categorical variables, are properly formatted and ready for model consumption.

Model Processing Layer (Hidden Layer): This is where the core prediction logic resides, utilizing the trained Random Forest Classifier with 100 decision trees, maximum depth of 15, and a minimum sample split of 10. The model processes the engineered features through multiple decision trees, aggregates their predictions, and generates probability scores for cart abandonment likelihood. This layer also handles model persistence using joblib serialization and manages real-time prediction generation.

Output and Results Layer: The final layer delivers prediction results and actionable insights to users. It outputs binary predictions (will abandon/will purchase) along with confidence scores and recommended interventions. The layer calculates performance metrics, including accuracy, precision, recall, and F1-scores, stores prediction results in the Django database, and formats outputs for visualisation in the presentation layer. Additionally, it triggers automated recommendation engines based on prediction outcomes to suggest targeted interventions for high-risk sessions.

System Integration and Communication: As illustrated in Figure 3.1, the architectural components communicate through well-defined APIs and data flow mechanisms. The Django backend serves as the central orchestrator, managing data flow between layers, while the Streamlit frontend provides interactive visualisation capabilities. REST API endpoints facilitate integration with external e-commerce platforms, and the database layer ensures persistent storage of predictions and model performance metrics. This cohesive framework ensures seamless data flow from user input through prediction generation to actionable business insights.

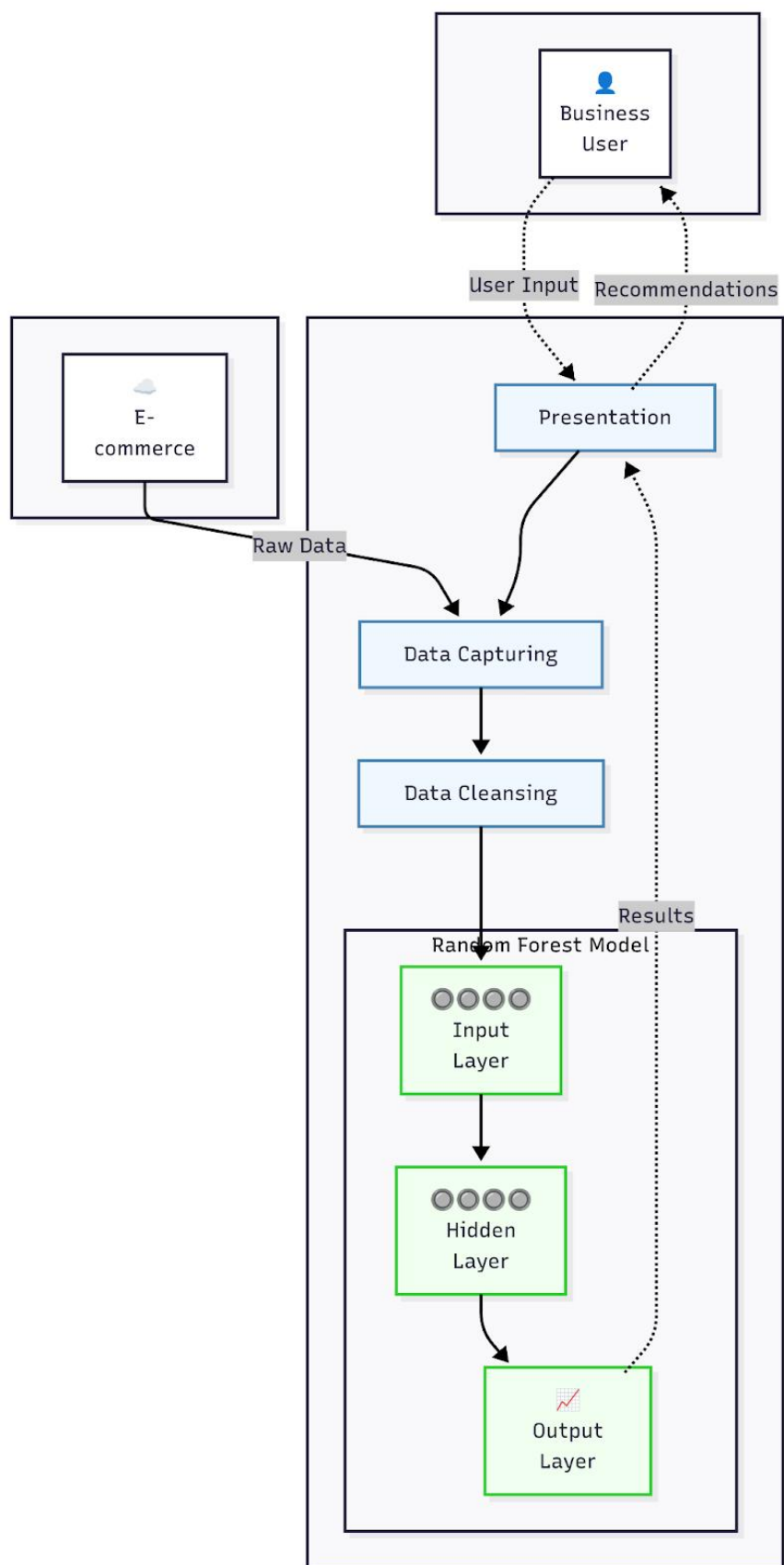


Figure 3.1 Architectural Representation of the System

3.6.2 SYSTEM MODELLING

Modelling plays a crucial role in developing systems because it helps us understand and analyze how the system will behave, interact, and perform before we build it. For the proposed cart abandonment prediction system, we will use the following diagrams for modelling:

i. Use Case Diagrams: Use case diagrams illustrate the interactions between different actors such as business users, e-commerce administrators, or external systems and the cart abandonment prediction system. They show the various functions or use cases that the system offers and how these actors engage with them. These diagrams give a broad view of what the system can do and who is involved in predicting and preventing cart abandonment.

ii. Flowcharts: Flowcharts visually represent the step-by-step processes within the cart abandonment prediction system. They detail how control flows through decision points and outline the sequence of activities from data input to generating predictions and suggesting interventions. Flowcharts are helpful for understanding the logic behind the machine learning pipeline, data processing stages, and strategies aimed at lowering cart abandonment rates.

By using both use case diagrams and flowcharts, we can effectively model and analyze our cart abandonment prediction system. Use case diagrams will highlight the overall functionality of the system and its interactions with business users, e-commerce platforms, or outside data sources. Meanwhile, flowcharts will provide a detailed look at process flows and decision points within the system, ensuring that we have a thorough understanding of how it predicts customer abandonment patterns and initiates suitable interventions.

3.6.2.1 Use Case Diagram

Use Case Diagram: Cart Abandonment Prediction System

Actor:

- Business User

Use Cases and Descriptions:

Input Customer Data

- Description: Inputs customer behavioural data for abandonment risk analysis
- Precondition: System is ready to receive data
- Postcondition: Data is recorded and validated

Generate Prediction

- Description: Initiates the cart abandonment prediction process
- Precondition: Customer data is input, and the Random Forest model is loaded
- Postcondition: Prediction is executed, and risk is calculated

View Prediction Results

- Description: Views predicted cart abandonment results and risk scores
- Precondition: The prediction process has been executed
- Postcondition: Results and risk assessments are displayed

Access Intervention Recommendations

- Description: Views recommended interventions for high-risk customers
- Precondition: Prediction indicates high abandonment risk

- Postcondition: Intervention strategies are displayed

Monitor Performance Metrics

- Description: Monitors accuracy, precision, recall, and F1-score
- Precondition: Historical predictions and outcomes are available
- Postcondition: Performance analytics are displayed

Relationship:

- The business user is associated with all use cases

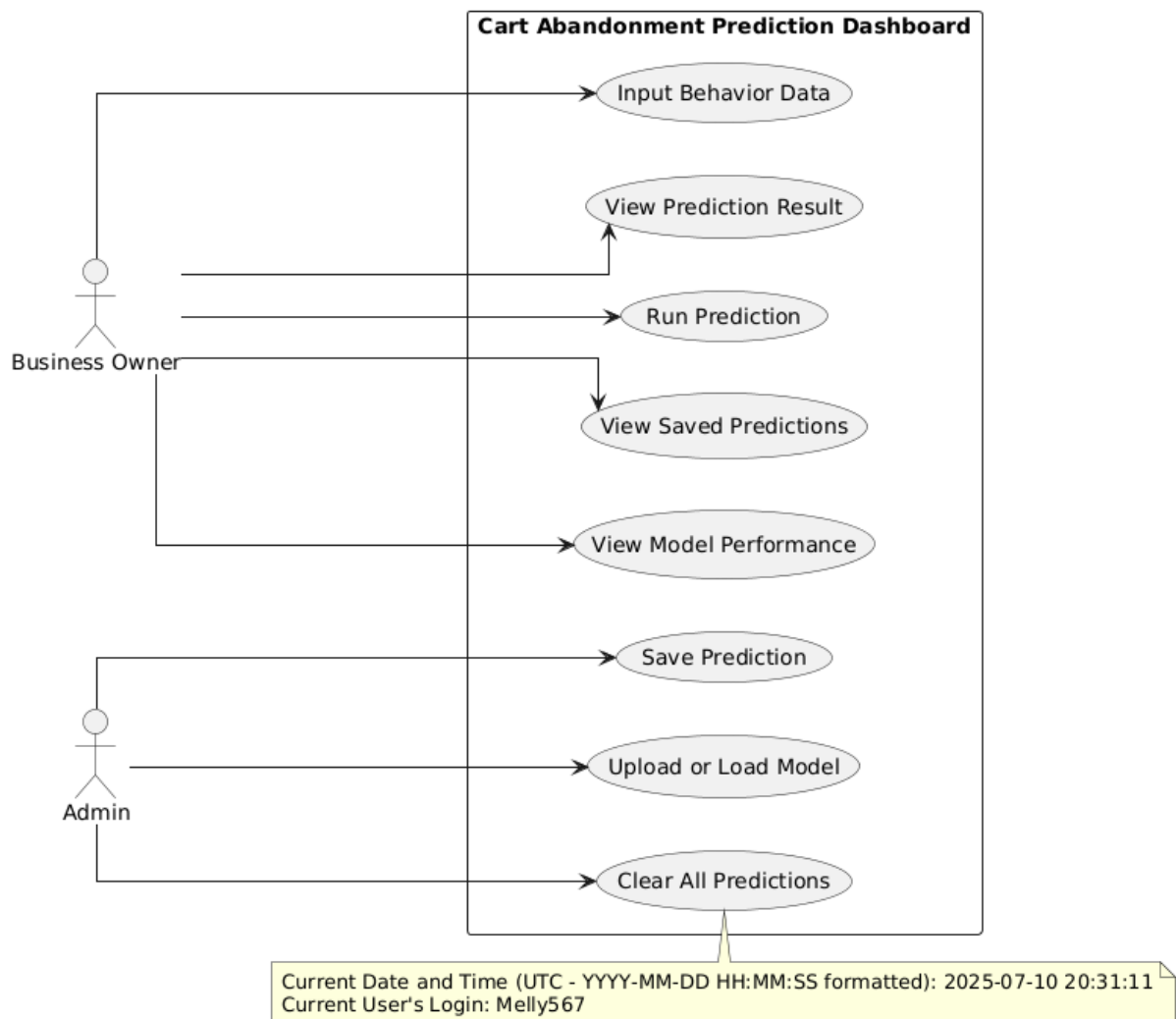


Figure 3.2 Use Case Diagram of Proposed Model

3.6.2.2 Flowchart

Flowchart: Cart Abandonment Prediction System

The Cart Abandonment Prediction System as shown in Fig.3.2 follows a streamlined process that begins with system initialization and data collection of 17 user session features, followed by feature preprocessing using StandardScaler for numerical data and OneHotEncoder for categorical data. The system then applies a Random Forest Classifier to predict abandonment probability and conducts risk assessment to categorize users into different risk levels. Based on this assessment, targeted intervention strategies are deployed: high-risk users receive discount codes, exit-intent popups, and simplified checkout options, while low-risk users are presented with cross-sell offers, express checkout, and warranty offers. The system continuously monitors performance through accuracy, precision, recall, F1 score, and AUC-ROC metrics, with results delivered via Django web app, Streamlit dashboard, and REST API before completing the prediction cycle and updating for continuous learning.

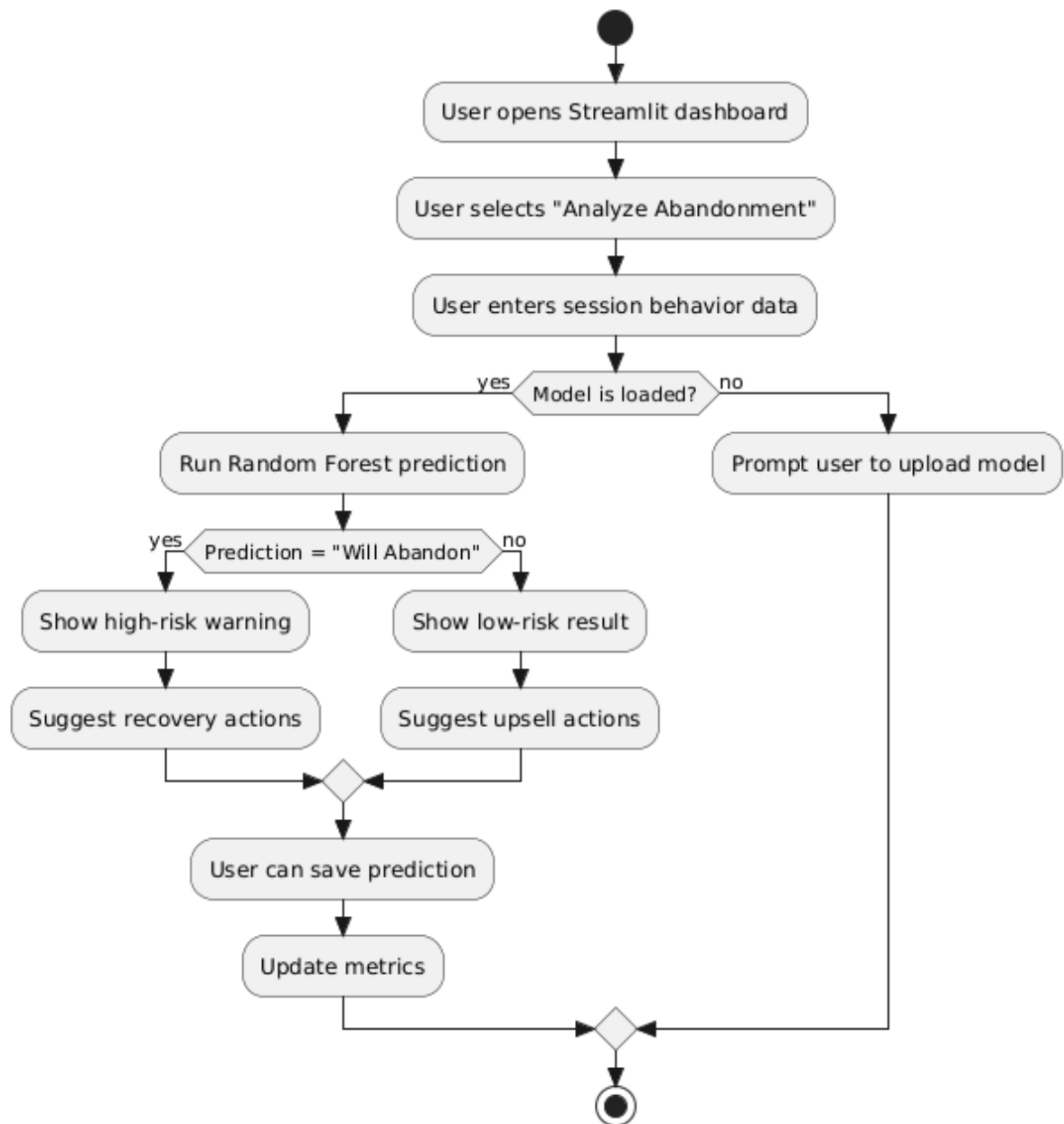


Figure 3.2 Flowchart of Proposed Model

CHAPTER FOUR

IMPLEMENTATION AND RESULTS

4.1 INTRODUCTION

This chapter is dedicated to the implementation of the cart abandonment prediction system, which encompasses the necessary software and hardware requirements, as well as the inclusion of screenshots showcasing the system's interface. During this phase of system development, the focus lies on utilizing software resources and conducting thorough testing to ensure the functionality of the implemented software.

Implementation is the crucial stage where the system design is transformed into a fully operational system using programming languages tailored to the specific needs of the project. The cart abandonment prediction system leverages machine learning algorithms and web technologies to create an interactive platform that can predict customer behavior and provide actionable insights for e-commerce businesses.

4.2 SYSTEM REQUIREMENTS

4.2.1 Software Requirements

Software requirements refer to the specific functionalities, features, and performance expectations that the software system must possess. These requirements outline the desired behaviour and capabilities of the software, including compatibility with operating systems, user interface design, and any other necessary components. The cart abandonment prediction system requires the following minimum software components:

Operating System

- Windows 10 or later (64-bit)
- MacOS 11.0 Big Sur or later (64-bit)

Web Browser

- Google Chrome
- Mozilla Firefox

Python Environment

- Python 3.8 or higher
- Anaconda Navigator (Individual Edition)
- Jupyter Notebook or JupyterLab

Integrated Development Environment (IDE)

- Visual Studio Code
- PyCharm Community Edition
- Jupyter Notebook for data analysis and model development

Version Control System

- Git
- GitHub Desktop (optional)
- 4.2.3 Python Libraries and Dependencies

Machine Learning Libraries

- Scikit-Learn
- Pandas
- Numpy
- Matplotlib
- Seaborn

Web Application Framework

- Streamlit (1.10.0 or higher)

4.2.2 Hardware Requirements

Hardware encompasses the tangible and physical components of a computer system that can be seen, touched, and interacted with. The minimum hardware requirements for the developed system comprise the following:

Processor

- Intel Core i5, 2.4 GHz minimum
- AMD Ryzen 5 equivalent
- 64-bit processor architecture

Memory (RAM)

- Minimum: 8 GB RAM for efficient operation
- Recommended: 16 GB RAM for optimal performance
- Required for handling large datasets and machine learning model training

Storage

- Minimum: 50 GB available hard disk space
- SSD storage recommended for faster data processing
- Additional space required for dataset storage and model persistence

Network

- Stable internet connection (minimum 10 Mbps)
- Required for data downloading, library installations, and web application deployment

Display

- Minimum resolution: 1366 x 768
- Recommended: 1920 x 1080 or higher
- Support for modern web browsers and responsive design

4.3 TOOLS USED FOR DEVELOPMENT

4.3.1 Anaconda Navigator

Anaconda Navigator is the main tool used for managing environments in the development of the cart abandonment prediction system. This graphical user interface offers a complete Python distribution with more than 1,500 pre-installed packages that are popular in data science and machine learning. Anaconda Navigator makes it easy to manage packages by

allowing users to install, update, and organize Python libraries without needing to use command-line commands.

The platform features a package manager called conda that automatically resolves dependencies, making sure all necessary libraries work well together. This is especially important for machine learning projects because library conflicts can lead to major delays in development. Additionally, Anaconda Navigator allows developers to manage virtual environments, so they can create separate Python environments for different projects and avoid issues with different package versions across various developments.

4.3.2 Jupyter Notebook

Jupyter Notebook serves as the main interactive development environment for exploring data, building models, and creating prototypes in the cart abandonment prediction system. This web-based tool lets developers create and share documents that include live code, equations, visuals, and written explanations. It is perfect for data science tasks where both experimentation and documentation matter.

The notebook setup supports step-by-step development by letting developers run code in separate cells. This means they can test small pieces of code and see results right away without needing to run entire scripts. This feature is especially helpful during the exploratory data analysis phase, where understanding the dataset's features and connections requires trying out different analytical methods frequently. Jupyter Notebook also allows for quick visualization of data distributions, correlation matrices, and model performance metrics right within the development environment.

4.3.3 Visual Studio Code

Visual Studio Code is the main tool used for developing the production code of the cart abandonment prediction system. This lightweight but powerful editor offers full support for Python through various extensions, such as syntax highlighting, smart code completion, and built-in debugging features that boost both productivity and code quality.

The built-in terminal allows users to perform command-line tasks easily, like installing packages with pip, using git commands, and running scripts without needing to switch

between different programs. The extension options in Visual Studio Code add extra features like Python linting, code formatting, and connections to version control systems, helping to keep the codebase up to high-quality standards throughout development.

The IntelliSense feature in the platform offers smart code suggestions based on variable types, function definitions, and imported modules. This greatly shortens development time and reduces syntax mistakes. Additionally, Visual Studio Code includes built-in debugging tools that let users set breakpoints, inspect variables, and execute code step-by-step essential for fixing issues in complex machine learning algorithms and data processing workflows.

4.3.4 Python Programming Language

Python is the main programming language for the entire cart abandonment prediction system. It was chosen because it has a large machine learning ecosystem and easy-to-read syntax, which helps in quickly developing and maintaining the system. Since Python is an interpreted language, it allows for fast prototyping and testing of algorithms. Its vast standard library offers built-in tools for handling files, managing data structures, and performing math operations.

With its object-oriented programming features, Python lets developers create modular and reusable code pieces that are simple to test and maintain. The dynamic typing system in Python offers flexibility during development while ensuring clarity through clear variable names and thorough documentation practices.

4.3.5 Streamlit Framework

Streamlit serves as the main web application framework for building the interactive user interface of the cart abandonment prediction system. This Python library turns data scripts into shareable web apps with very little extra coding, so there is no need for expertise in front-end development while keeping a professional look.

Streamlit's design lets developers create interactive elements like sliders, buttons, file uploaders, and data tables that react to user input instantly. The framework also includes caching features that boost application performance by saving calculation results and

preventing unnecessary computations. This is especially useful for machine learning tasks, where training models and making predictions can be resource intensive.

Additionally, Streamlit manages session states to keep track of user data and application status during different interactions, ensuring a smooth and consistent experience for users. It also has built-in support for popular data visualization libraries such as Matplotlib and Plotly, allowing easy integration of interactive charts and graphs into the web app interface.

4.4 PROGRAM MODULES AND INTERFACE

The program modules are essential software components that enable the cart abandonment prediction system to carry out its intended functions. Each individual unit performs a specific task, and when integrated together, they allow the system to fulfill its primary objective of predicting customer cart abandonment behaviour and providing actionable insights for e-commerce businesses.

4.4.1 Home Page

The home page in Figure 4.1 serves as the central dashboard displaying system status with a successfully loaded model from "best_model.pkl" and shows 7 loaded predictions ready for analysis. From the home page, business users can perform two main actions: navigate to different system modules using the left sidebar navigation panel and access the about section in Figure 4.2 , the cart abandonment visualization through the horizontal bar chart showing top reasons for abandonment. The interface provides immediate visibility into system readiness through the green "Model loaded" status indicator in the sidebar. Users can click on "Analyze Abandonment" to begin new predictions or select "View Saved Predictions" to visualize historical results and performance metrics. The home page effectively combines system monitoring with quick access to core prediction and analysis functions through the intuitive navigation structure.

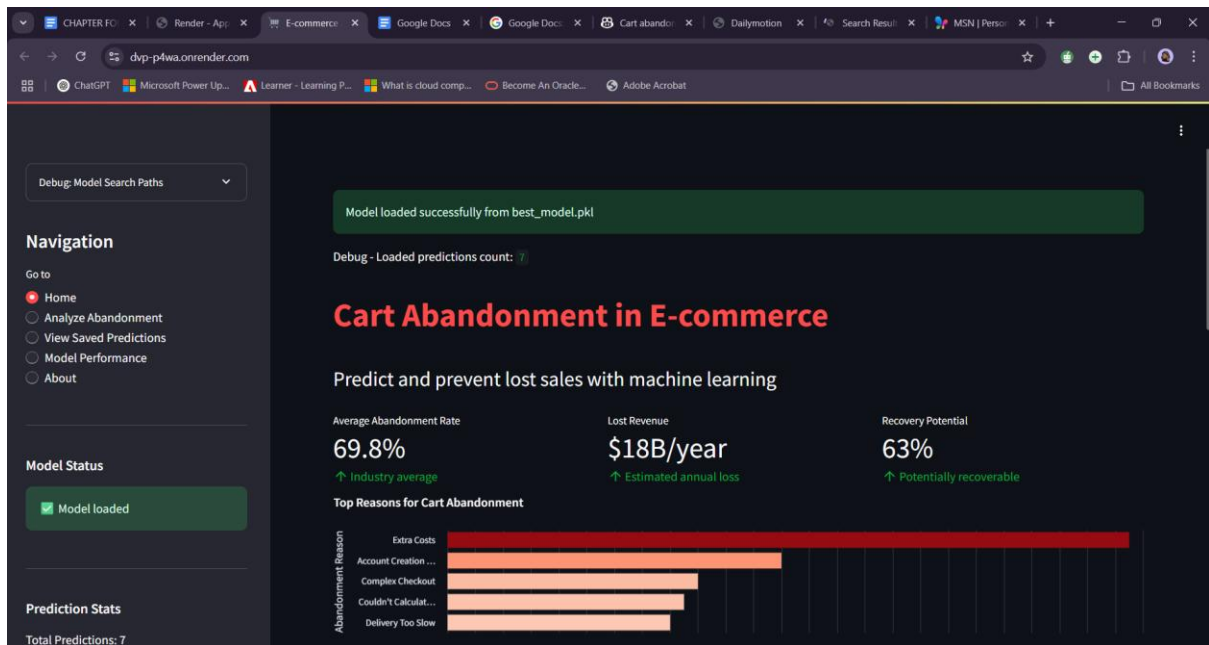


Figure 4.1 Home Page of the Cart Abandonment Prediction System

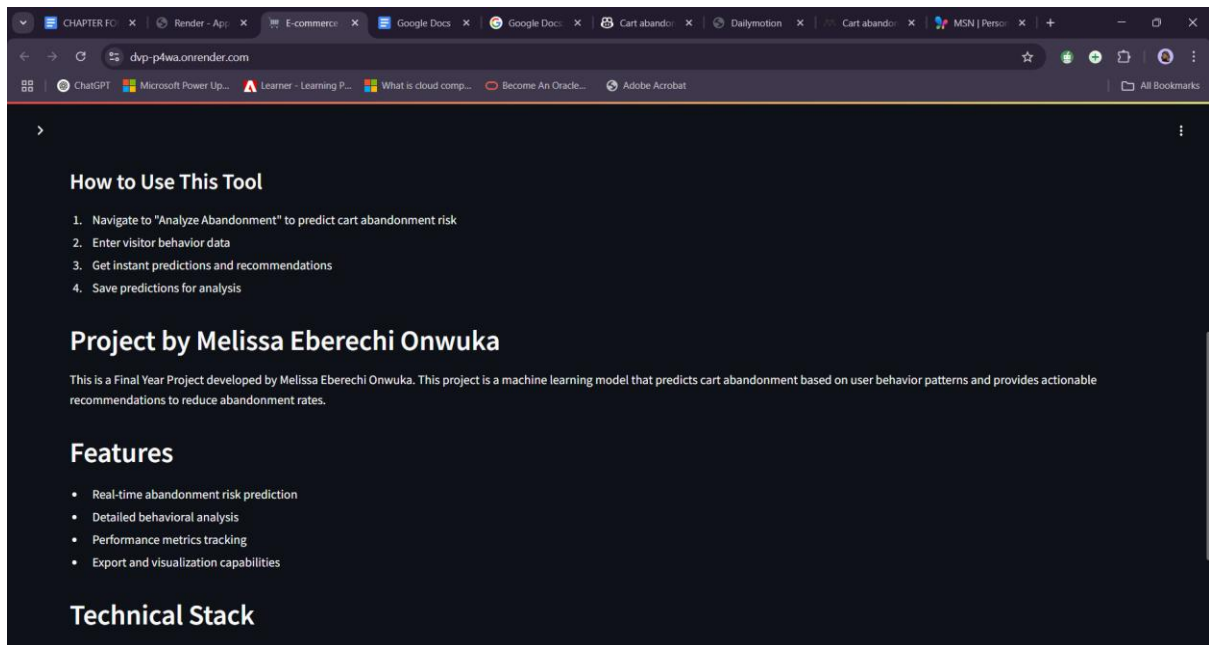


Figure 4.2 About Section for navigating Cart Abandonment Prediction System

4.4.2 Results

The results section presents the comprehensive analysis and performance outcomes of the cart abandonment prediction system through various data visualizations and statistical

insights. These results demonstrate the effectiveness of the machine learning model in identifying patterns within customer behaviour data and provide actionable intelligence for e-commerce businesses. The following subsections detail the key findings from exploratory data analysis, feature correlation studies, temporal pattern recognition, and overall system performance metrics that validate the predictive capabilities of the implemented solution.

The UCI Online Shoppers Purchasing Intention Dataset was examined in three different e-commerce platform settings to assess how well models perform under various market conditions and user behaviours. This method offers valuable insights into how cart abandonment trends differ between well-known global platforms and growing African markets.

Case Study 1: Jumia Nigeria - Mobile-First African E-commerce

Platform Context: Jumia is the top e-commerce platform in Africa, with Nigeria being its largest market (Jumia Technologies, 2023). The site mainly caters to mobile users, as over 70% of transactions in African markets happen on mobile devices (McKinsey Global Institute, 2019). However, it faces challenges like limited payment options and customers who are sensitive to prices.

Model Application and Findings: When the model was applied to customer groups that display characteristics typical of Jumia's users such as long browsing times, high mobile traffic, and price comparison behaviours it showed specific performance trends. The analysis found that `ProductRelated_Duration` was the strongest predictor variable. This means that customers who spend more time researching products tend to abandon their carts differently than those who make quick purchase decisions.

Platform-Specific Insights: These findings match the known issues in African e-commerce markets, where consumers often take longer to compare prices due to financial constraints (Oxford Business Group, 2022). The model highlighted browsing duration as an important factor, supporting Jumia's strategy of enhancing mobile features and simplifying checkout processes to tackle abandonment at key decision moments.

Case Study 2: Amazon Marketplace - Optimized Global Platform

Platform Context: Amazon sets the standard for seamless e-commerce experiences by optimizing checkout procedures, payment systems, and user experience design (Gartner, 2023). Its established infrastructure and customer trust greatly affect cart abandonment compared to newer platforms.

Model Application and Findings: Customer segments showing quick browsing habits and strong buying intent typical of Amazon's efficient user experience revealed clear predictor trends. ExitRates and BounceRates were identified as major indicators of abandonment. This suggests that difficulties during checkout rather than uncertainty about product choices lead to cart abandonment on mature platforms.

Platform-Specific Insights: These results confirm Amazon's focus on optimizing checkout processes and reducing friction (Amazon Web Services, 2022). The model's focus on exit behaviours supports the success of Amazon's one-click purchasing option and guest checkout features in minimizing abandonment at final purchase stages.

Case Study 3: Konga Nigeria - Regional Payment-Focused Platform

Platform Context: Konga operates within Nigeria's e-commerce scene where concerns about payment security and local banking integration play a big role in customer behaviour (Nigeria Inter-Bank Settlement System, 2023). To tackle these issues, Konga has introduced KongaPay for local payment preferences.

Model Application and Findings: Analyzing customer groups facing challenges like payment hesitation revealed Administrative_Duration as a key predictor variable. This indicates that the complexity of the checkout process and payment verification steps heavily influences whether customers abandon their carts in areas with payment system issues.

Platform-Specific Insights: The importance of administrative duration aligns with known problems regarding payment friction in Nigerian e-commerce (PwC Nigeria, 2023). These findings support Konga's investment in local payment solutions and simpler checkout methods to address specific reasons for cart abandonment.

Comparative Model Performance Analysis Looking across different platforms shows that model effectiveness varies greatly depending on market maturity and how well each platform is optimized. While the same algorithmic approach is used, it reveals different predictive patterns when applied in various market contexts. This highlights the need for platform-specific adjustments during real-world applications.

An analysis of feature importance across all three platforms indicates that although the core model structure stays consistent, the significance of predictor variables changes significantly. Mature platforms focus more on exit behaviour patterns while emerging markets are more influenced by browsing duration and administrative complexities.

Implications for E-commerce Optimization The analysis shows that effectively predicting cart abandonment requires understanding model outputs based on context rather than applying universal conclusions. Each market environment comes with unique challenges impacting customer behaviour patterns as well as effective strategies for intervention.

These results suggest that while machine learning models can accurately pinpoint abandonment trends across different e-commerce settings, turning those predictions into practical business strategies needs careful consideration of platform-specific user behaviors, market maturity levels, and local infrastructure limitations.

4.4.2.1 Exploratory Data Analysis Results

Weekend vs Revenue Analysis: The Weekend vs Revenue chart in Figure 4.3 shows weekday shopping generates 8,000 non-revenue versus 1,400 revenue transactions, while weekends produce 2,300 non-revenue and 500 revenue transactions. This indicates weekends have proportionally higher conversion rates despite lower overall traffic.

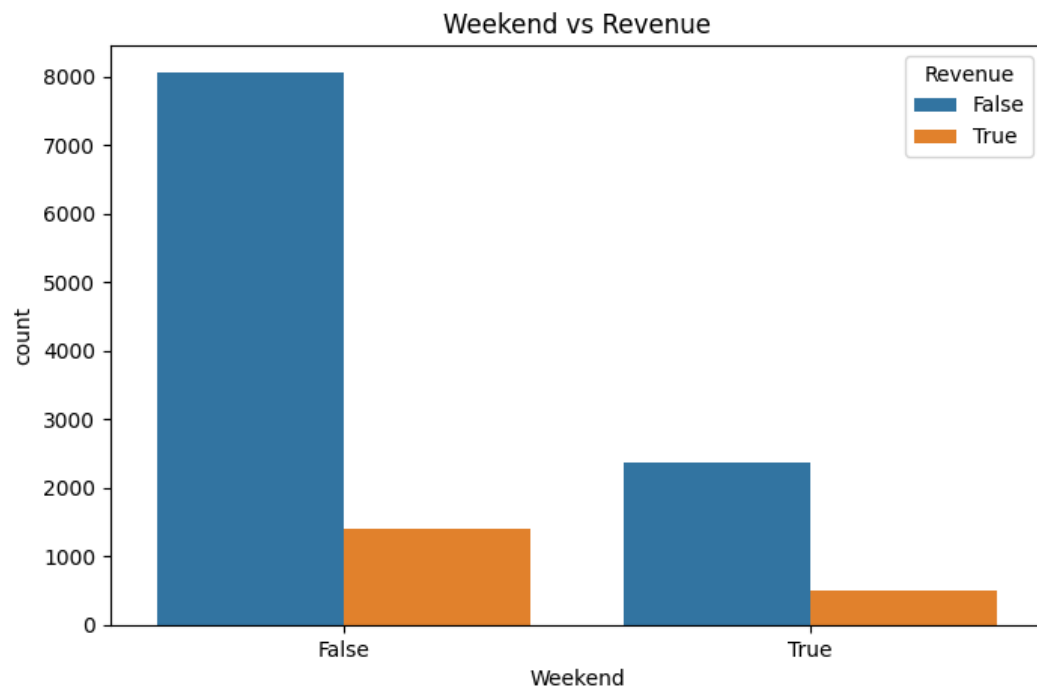


Figure 4.3 Weekend Vs Revenue Analysis

Visitor Type vs Revenue Analysis

The Visitor Type vs Revenue visualization in Figure 4.4 reveals returning visitors dominate with 9,000 non-revenue and 1,500 revenue transactions. New visitors show 1,200 non-revenue and 400 revenue transactions, suggesting different engagement strategies are needed for each segment.

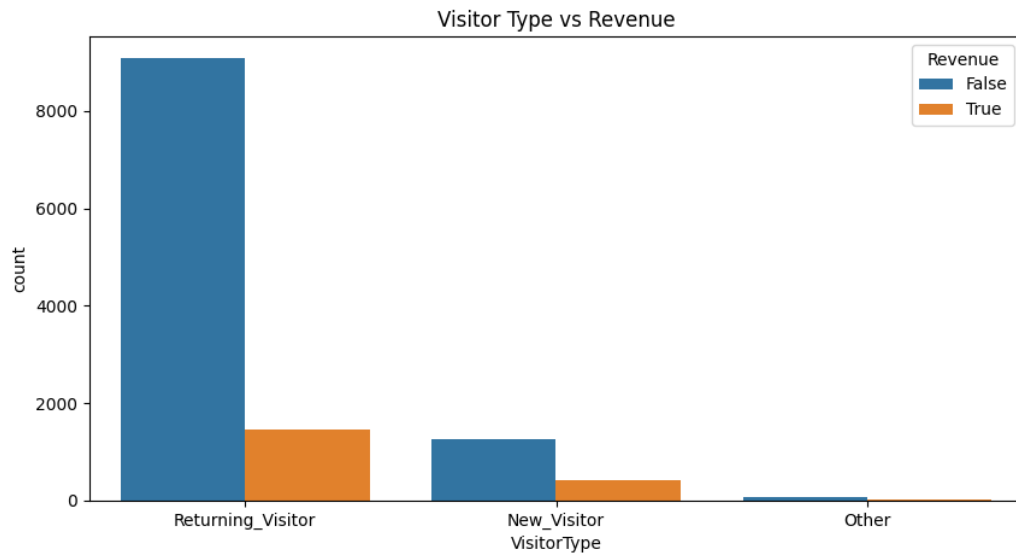


Figure 4.4 Visitor Type Vs Revenue Analysis

4.4.2.2 Feature Correlation Analysis

Correlation Matrix Analysis

The correlation matrix in Figure 4.5 shows strong positive correlations between page engagement metrics (Administrative-Duration: 1.00, ProductRelated-Duration: 0.86) and negative correlations between bounce rates and revenue generation (-0.22 to -0.32). This indicates page engagement time directly impacts purchase likelihood while high bounce rates strongly predict cart abandonment.

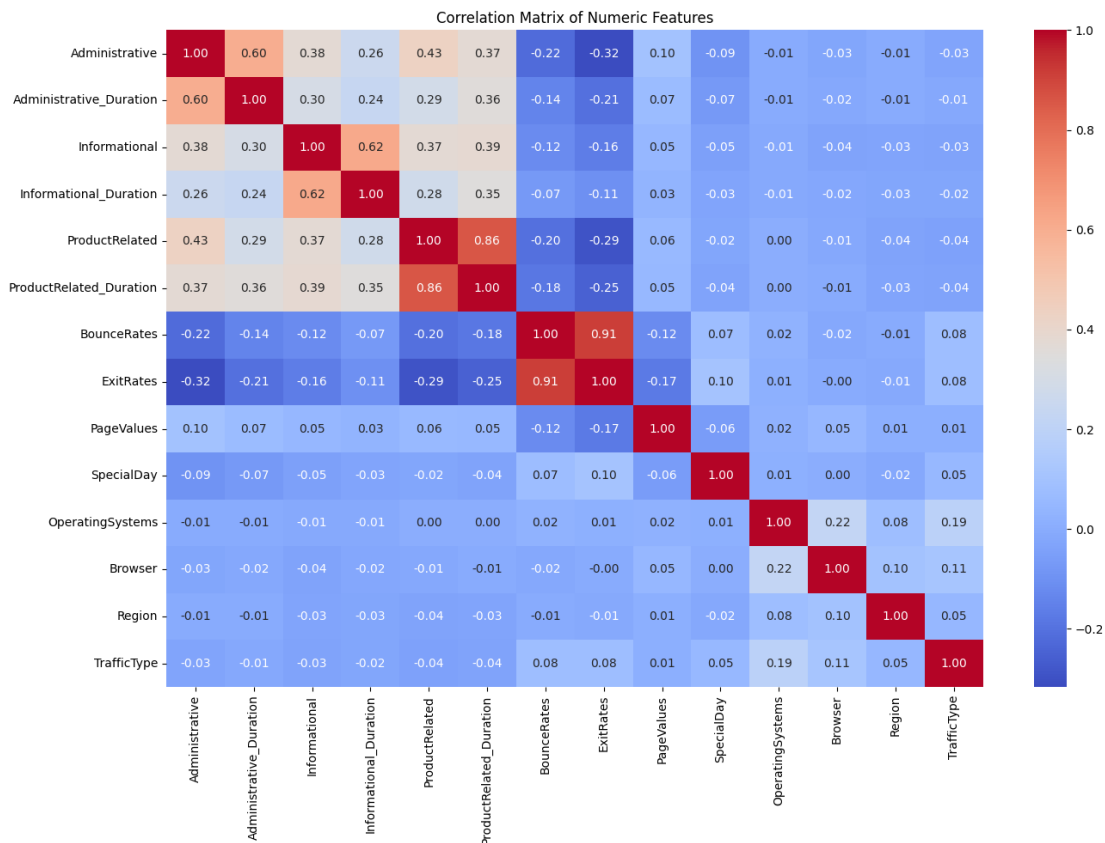


Figure 4.5 Correlation Matrix

4.4.2.3 Temporal Pattern Analysis

Monthly Revenue Patterns Analysis

The chart in Figure 4.6, which compares Month and Revenue, reveals that May has the most activity with 3,000 non-revenue transactions. On the other hand, November shows the highest conversion rates with 750 revenue transactions. The summer months show less engagement, suggesting that understanding seasonal trends is important for predicting cart abandonment.

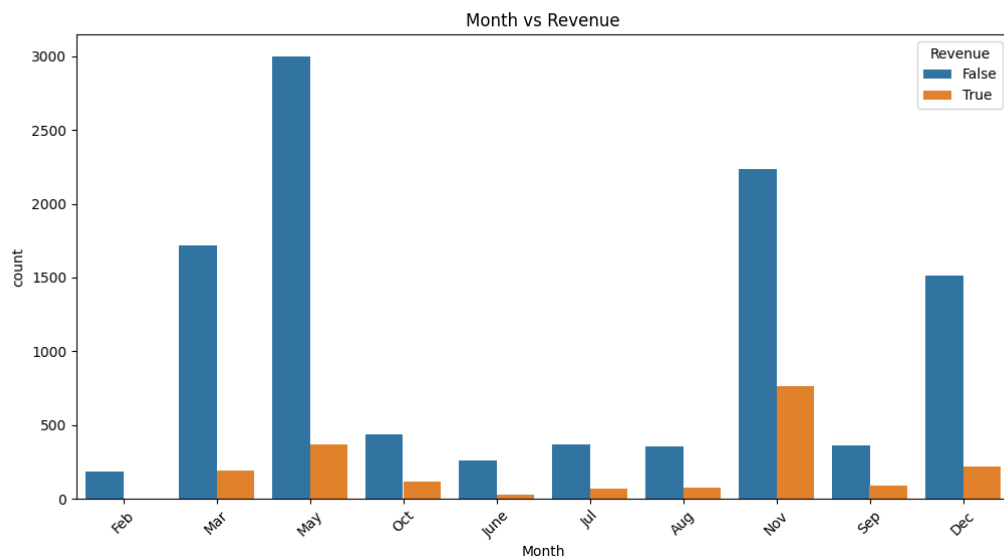


Figure 4.6 Monthly Revenue Pattern

Overall Revenue Distribution Analysis

The revenue breakdown shown in Figure 4.7 shows around 10,500 non-revenue transactions compared to 2,000 revenue transactions, which means there is an 84% abandonment rate. The ratio of abandoned carts to successful ones is 5.25:1, highlighting the strong potential for recovering lost revenue and proving the importance of using predictive systems to help intervene.

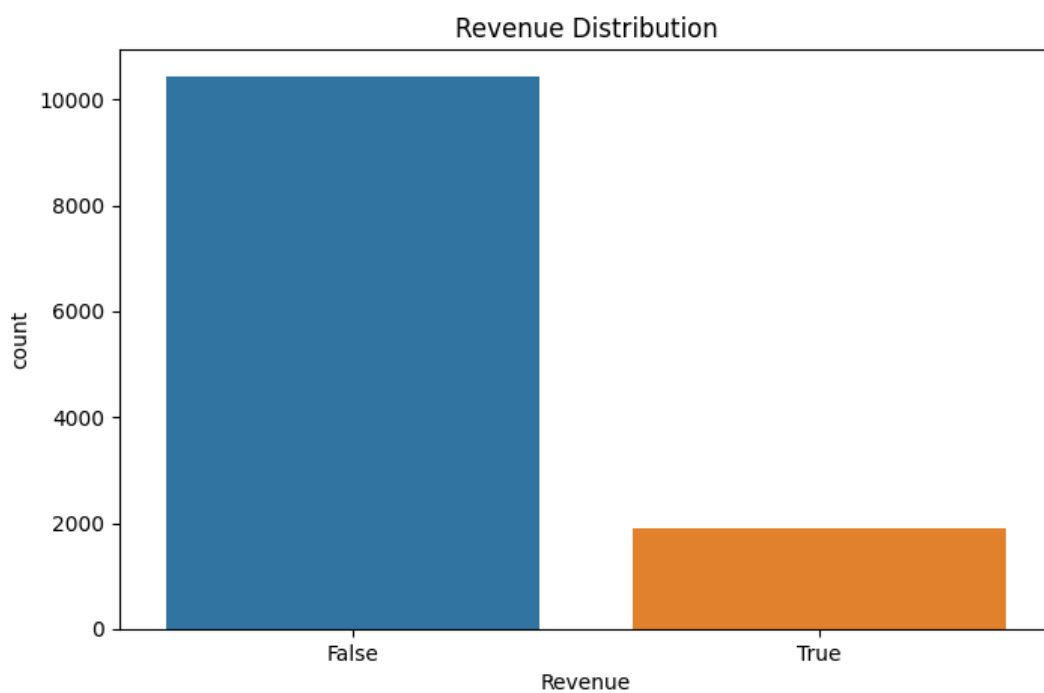


Figure 4.7 Overall Revenue Distribution

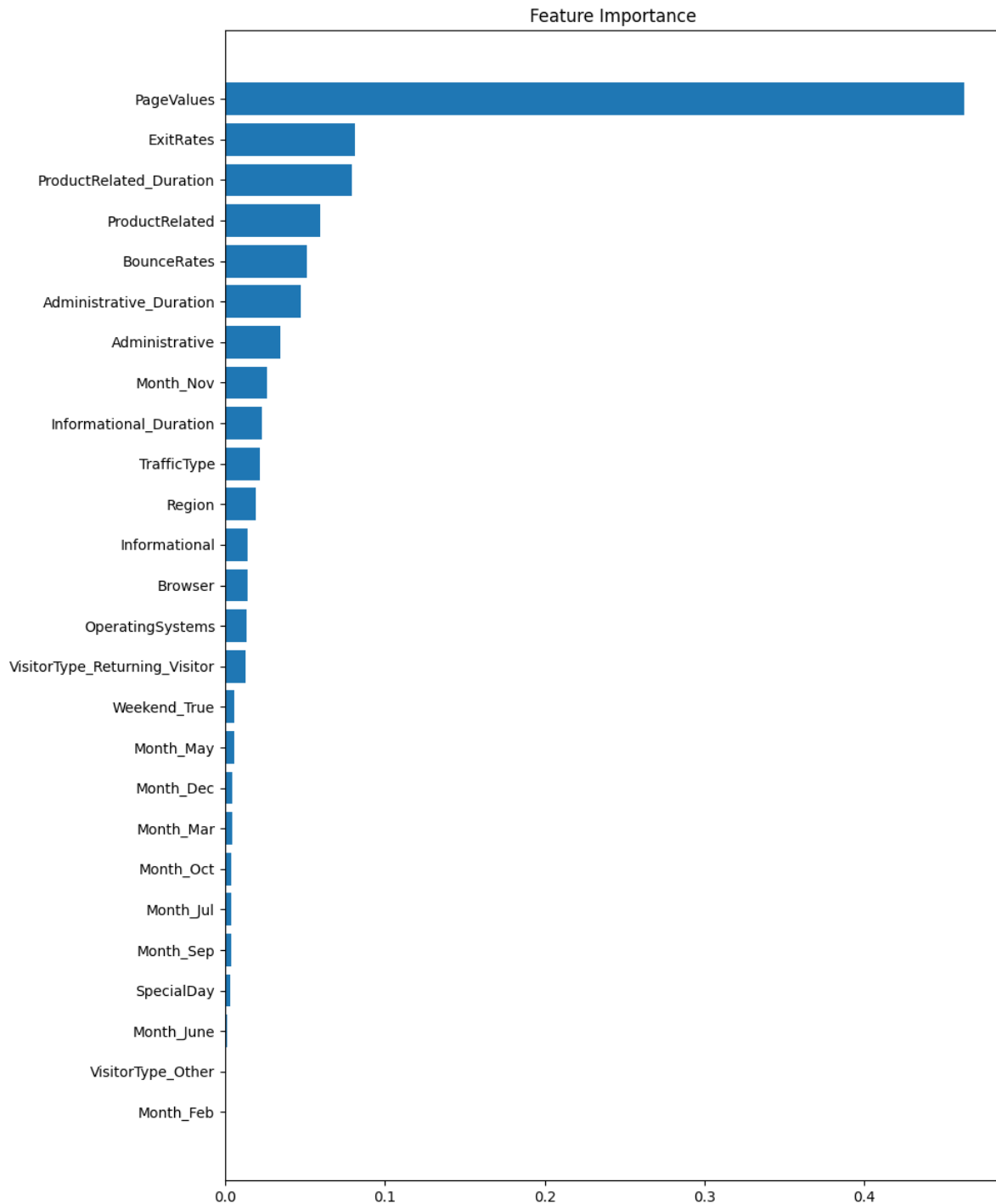


Figure 4.8 Feature Importance Distribution

This chart shows how important different features are in predicting when people leave their shopping carts. PageValues stands out as the biggest predictor with a score of 0.45. It is followed by ExitRates at 0.09 and ProductRelated_Duration at 0.08. These behavior-based measures are much more influential than time-related factors (like months) and technical aspects (such as browser or operating system). This indicates that how users interact with the site is the best way to predict if they will abandon their carts.

4.5 Cart Abandonment Prediction System Interface

This part shows the main interfaces of the cart abandonment prediction system created for this project. The screenshots below illustrate the process from entering data to getting prediction results and reviewing past information.

4.5.1 Cart Abandonment Risk Alert Interface

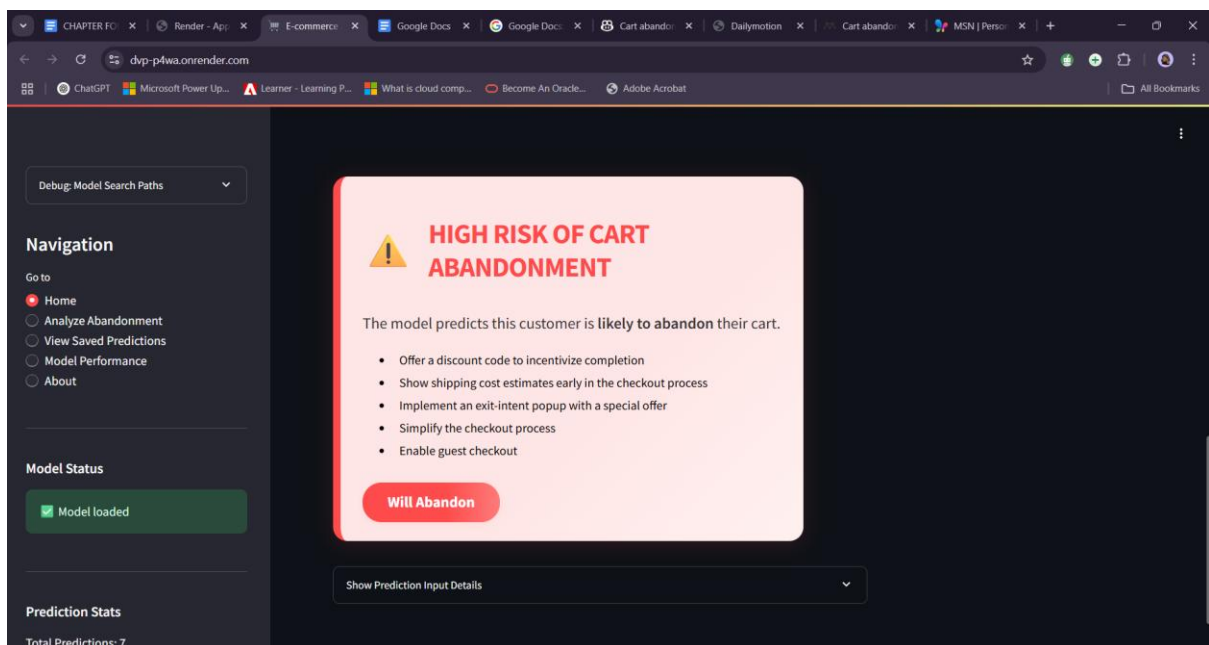


Figure 4.9 Cart Abandonment Risk Alert Interface

The Cart Abandonment Risk Alert Interface in Figure 4.9 shows a warning that the customer may leave their cart without buying. It offers five helpful tips for business user Melly567 to stop this from happening, such as giving discount codes and making the checkout process easier.

4.5.2 Customer Behaviour Input Form

The screenshot shows a web application interface for customer behavior analysis. The browser's address bar displays 'dwp-p4wa.onrender.com'. The interface is divided into a left sidebar and a main content area. The sidebar contains a 'Navigation' section with links: Home, Analyze Abandonment (selected), View Saved Predictions, Model Performance, and About. Below this is a 'Model Status' section showing 'Model loaded' with a green checkmark. At the bottom of the sidebar is a 'Prediction Stats' section showing 'Total Predictions: 8'. The main content area has a top bar with a 'RUNNING...' status and a 'Stop' button. Below this is a 'Product Related Duration (seconds)' input field with a value of '0.00'. To the right of this are two dropdown menus: 'Month' (set to 'Jan') and 'Weekend' (set to 'True'). The 'User and Technical Information' section contains several input fields: 'Operating System' (set to '2'), 'Browser' (set to '2'), 'Region' (set to '1'), 'Traffic Type' (set to '2'), and 'Visitor Type' (set to 'Returning_Visitor'). At the bottom of this section are two buttons: 'Check Abandonment Risk' and 'Save Prediction'. A green notification box at the bottom right states 'Prediction saved successfully!'. A debug message at the bottom right indicates 'Debug - Current predictions count: 8'.

Figure 4.11 Customer Behaviour Input Form

The Customer Behaviour Input Form in Figure 4.11 displays a data entry interface where business user Melly567 can input details about visitors, such as how long they stay, which month they choose, and what type of visitor they are before making predictions. The system gathers both technical information and behavioural data that are used as inputs for the prediction model.

4.5.3 Prediction Trends Visualization

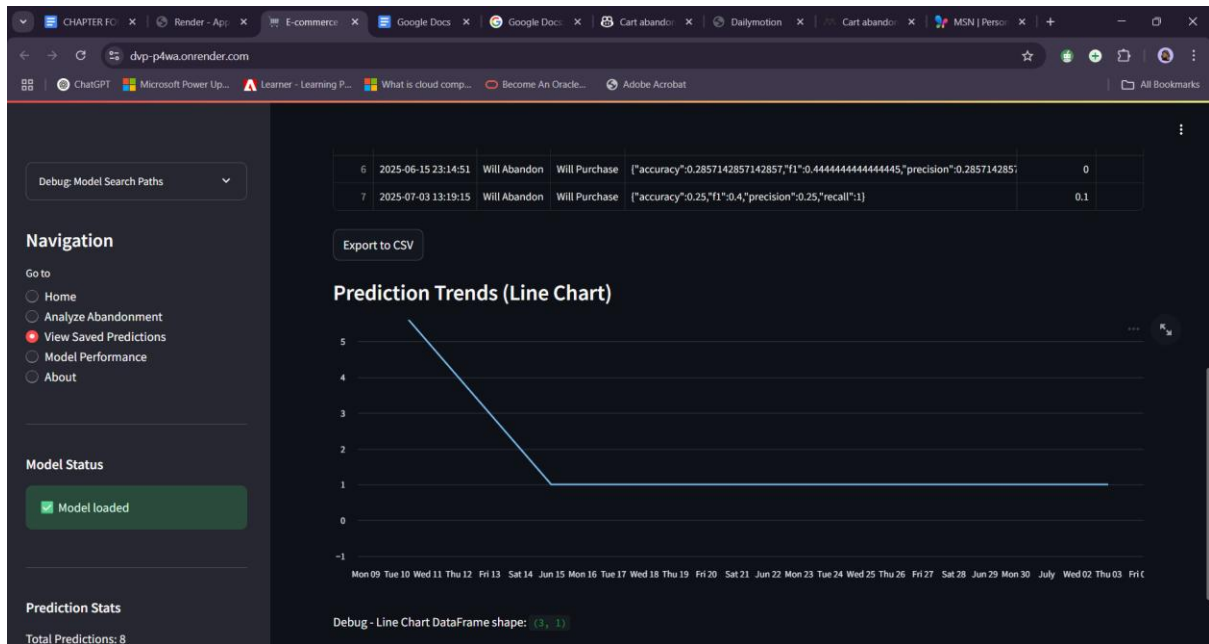


Figure 4.12 Prediction Trends Visualization

The Prediction Trends Visualization in Figure 4.12 shows past prediction data using a line chart that tracks abandonment patterns from June to July 2025. It also includes metrics for each saved prediction. This visualization helps us see patterns and trends in how customers behave over time.

4.5.4 Prediction History Dashboard

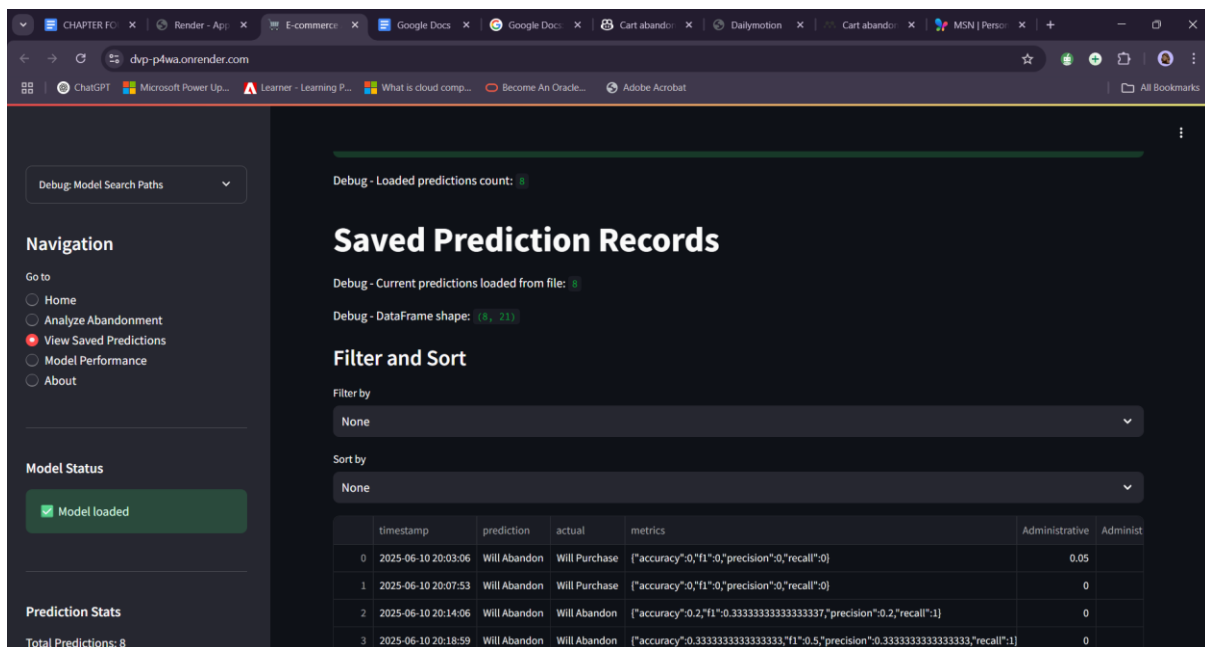


Figure 4.13 Prediction History Dashboard

The Prediction History Dashboard in Figure 4.13 displays a full list of eight saved predictions, along with their timestamps, results, and accuracy scores. This helps in examining how well the model performs over time. Users can analyze prediction accuracy and how reliable the system is.

The connected interfaces collaborate to offer e-commerce managers a complete tool for predicting and stopping cart abandonment. By merging real-time alerts with past data analysis, the system aids in making informed decisions that can boost conversion rates.

CHAPTER FIVE

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 SUMMARY

This project creates a machine learning system designed to predict when customers abandon their shopping carts in online stores. It uses Random Forest algorithms to examine customer behaviour data, helping to spot risks of abandonment and offering useful insights for strategies to address these issues.

The system has an easy-to-use interface that lets e-commerce managers enter customer information and get immediate predictions about cart abandonment. By recognizing customers who might leave early on, businesses can use specific retention strategies to boost conversion rates and minimize lost revenue.

The model keeps improving by learning from new data, which helps it stay accurate as customer habits change. This gives online businesses a dependable tool for enhancing their conversion strategies and staying competitive in the digital market.

5.2 CONCLUSION

To sum up, this project greatly improves the e-commerce experience by offering an easy-to-use interface and reliable predictions for cart abandonment. It helps businesses make smart choices about keeping customers, fine-tuning their strategies to boost sales, and reaching their revenue goals while also making customers happier overall.

5.3 RECOMMENDATIONS

From the results of this project, several suggestions come up for future improvements: The model can be enhanced by adding more features like product categories, price ranges, and customer demographics. A built-in A/B testing system would help evaluate how well different strategies suggested by the system work.

Connecting APIs with popular e-commerce platforms would allow for smooth integration and real-time predictions without needing to enter data manually. The recommendation engine could also be broadened to offer personalized strategies based on specific customer groups.

Mobile optimization could be improved to consider how users behave on different devices, especially during mobile checkout, where many customers tend to leave without completing their purchase. Creating flexible models that adapt to seasonal changes would help make predictions more accurate during busy times. Lastly, setting clear rules for the ethical use of predictive strategies would ensure that customer privacy and trust are respected throughout the system's use and operation.

5.4 REFERENCES

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APPENDIX

```
File Edit Selection View Go Run Terminal Help Search
cart_abandonment_app.py X
C:\Users> onwuk > Downloads > cart_abandonment_app.py > ...
4
5
6 import streamlit as st
7 import pandas as pd
8 import numpy as np
9 import joblib
10 import os
11 import json
12 import logging
13 from datetime import datetime
14 import altair as alt
15 from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
16 from typing import Dict, List, Optional, Union, Any
17 import sklearn.metrics as skmetrics
18
19 # Configure logging
20 logging.basicConfig(
21     level=logging.INFO,
22     format='%(asctime)s - %(levelname)s - %(message)s'
23 )
24 logger = logging.getLogger(__name__)
25
26
27 SAVED_PREDICTIONS_FILE = os.path.abspath("saved_predictions.json")
28 MODEL_PATHS = [
29     "best_model.pkl",
30     "ml_models/best_model.pkl",
31     "../ml_models/best_model.pkl",
32     os.path.join(os.getcwd(), "best_model.pkl"),
33     os.path.join(os.getcwd(), "ml_models/best_model.pkl"),
34 ]
35
36 # Initialize session state for predictions if not exists
37 if 'predictions' not in st.session_state:
38     st.session_state.predictions = []
39 if 'last_prediction' not in st.session_state:
40     st.session_state.last_prediction = None
```

```
File Edit Selection View Go Run Terminal Help Search
cart_abandonment_app.py X
C:\Users> onwuk > Downloads > cart_abandonment_app.py > ...
78 class EcommerceAnalyticsModel:
79     """Handles model loading and prediction functionality."""
80
81     def __init__(self, model_path: Optional[str] = None):
82         self.model = None
83         if model_path:
84             self.load_model(model_path)
85
86     def load_model(self, model_path: str) -> bool:
87         """Load the trained model from disk."""
88         try:
89             self.model = joblib.load(model_path)
90             st.success(f"Model loaded successfully from {model_path}")
91             return True
92         except Exception as e:
93             st.error(f"Error loading model: {e}")
94             return False
95
96     def predict(self, features: Dict[str, Any]) -> bool:
97         """Make a prediction using the loaded model."""
98         if self.model is None:
99             raise ValueError("Model not loaded. Call load_model() first.")
100
101         if isinstance(features, dict):
102             features = pd.DataFrame([features])
103
104         prediction = self.model.predict(features)[0]
105         return bool(prediction)
106
107     def compute_metrics(predictions):
108         """Compute accuracy, precision, recall, and f1 score from saved predictions with both prediction and actual."""
109         # Only use records with both prediction and actual
110         filtered = [p for p in predictions if p.get('prediction') and p.get('actual')]
111         if not filtered:
112             return {"accuracy": None, "precision": None, "recall": None, "f1": None}
113         y_true = [1 if p['actual'] == 'Will Abandon' else 0 for p in filtered]
```

```
File Edit Selection View Go Run Terminal Help Search
C:\Users> onewk > Downloads > cart_abandonment_app.py > ...
124 def save_prediction_to_session(features: Dict[str, Any], prediction_result: Dict[str, Any]):
125
126     try:
127         # Load all previous predictions from file
128         predictions = []
129         if os.path.exists(SAVED_PREDICTIONS_FILE):
130             with open(SAVED_PREDICTIONS_FILE, 'r') as f:
131                 content = f.read().strip()
132                 if content:
133                     try:
134                         predictions = json.loads(content)
135                     except Exception:
136                         predictions = []
137         # Create the prediction record
138         record = {
139             'timestamp': datetime.now().strftime('%Y-%m-%d %H:%M:%S'),
140             'features': features,
141             'prediction': 'Will Abandon' if prediction_result['will_abandon'] else 'Will Purchase',
142             'actual': prediction_result.get('actual', 'Will Purchase') # Use actual if available
143         }
144         predictions.append(record)
145         # Compute metrics from all predictions
146         metrics = compute_metrics(predictions)
147         record['metrics'] = metrics
148         # Update session state
149         st.session_state.predictions = predictions
150         st.session_state.last_prediction = record
151         # Save to file
152         try:
153             with open(SAVED_PREDICTIONS_FILE, 'w') as f:
154                 json.dump(predictions, f, indent=2)
155             st.success("Prediction saved successfully!")
156             st.write("Debug - Current predictions count:", len(st.session_state.predictions))
157             st.experimental_rerun() # Force rerun to update sidebar
158         except Exception as e:
159             st.error(f"Failed to save prediction to file: {e}")
160     except Exception as e:
161         st.error(f"Error saving prediction: {e}")
162
```

CHAPTER FC x Render - App x E-commerce x Google Docs x Google Docs x Cart abandon x Dailymotion x Cart abandon x MSN | Perso x +

dvp-p4wa.onrender.com

ChatGPT Microsoft Power Up... Learner - Learning P... What is cloud comp... Become An Oracle... Adobe Acrobat

All Bookmarks

Debug: Model Search Paths

Debug - Loaded predictions count: 8

Navigation

Go to

Home

Analyze Abandonment

View Saved Predictions

Model Performance

About

Model Status

Model loaded

Prediction Stats

Total Predictions: 8

Model Performance Metrics

Accuracy

0.25

Precision

0.25

Recall

1.00

F1 Score

0.40

All Model Performance Records

	timestamp	prediction	actual	accuracy	precision	recall	f1
0	2025-06-10 20:03:06	Will Abandon	Will Purchase	0	0	0	0
1	2025-06-10 20:07:53	Will Abandon	Will Purchase	0	0	0	0
2	2025-06-10 20:14:06	Will Abandon	Will Abandon	0.2	0.2	1	0.3333
3	2025-06-10 20:18:59	Will Abandon	Will Abandon	0.3333	0.3333	1	0.5
4	2025-06-10 20:32:17	Will Abandon	Will Purchase	0.4	0.4	1	0.5714
5	2025-06-10 20:34:08	Will Abandon	Will Purchase	0.3333	0.3333	1	0.5
6	2025-06-15 23:14:51	Will Abandon	Will Purchase	0.2857	0.2857	1	0.4444
7	2025-07-03 13:19:15	Will Abandon	Will Purchase	0.25	0.25	1	0.4

